

Laboratory Manual

Rock Mechanics Practical (MEC 15201)

for

5th B.Tech (Mining Engineering)



**Department of Mining Engineering
Indian Institute of Technology (Indian School of Mines),
Dhanbad**

Course Type	Course Code	Name of Course	L	T	P	Credit
DC	MEC15201	Rock Mechanics Practical	0	0	2	2

Course Objective
The objective of this practical is to enable the learner to understand the determination of different physico-mechanical properties of the intact rock, soil and joint as per the suggested methods (ISRM, ASTM, BIS etc.) and also provide an overview of their application in ground control, mine excavations and geo-engineering design.
Learning Outcomes
Upon successful completion of this course, students will: - know determination of various physico-mechanical properties of rocks and soils and - correlate their application in ground control, mine excavations and geo-engineering design

SL. No.	Name of the Experiment	Practical Hours	Learning Outcome
1	Preparation of rock samples for testing	2	Physico-mechanical properties for rock characterization
2	Determination of uniaxial compressive strength and tensile strength of rock samples	2	Characterization and classification of rocks in design of structures, Strength and deformability of rocks in unconfined and confined conditions and their relation to mine design
3	Determination of modulus of elasticity and poisson's ratio using strain gauges	2	
4	Determination of point load strength index of rock samples	2	Strength index of rock for irregular rock samples; Quick estimation of compressive strength of rock for engineering classification and design
5	Determination of Protodyakonov strength index and slake durability index of rock	2	Strength index of rock for irregular rock samples; workability of coal seams and engineering classification of rocks; An index to alteration and relative ranking of rock durability and weatherability of rocks.
6	Determination of triaxial strength of rock samples	2	Strength and deformability (elastic properties) of rock for design of underground structures such as mine excavations, mine pillars, tunnels etc.
7	Determination of Cerchar Abrasivity Index of rocks	2	Rock abrasivity and its application in tool consumption estimation
8	Determination of strength of concrete/rock using Schmidt hammer	2	Strength criterion of rocks, relative strength of concrete/rocks insitu (portable) by non-destructive testing
9	Determination of In-situ stress measurement using hydro-fracturing technique	2	In-situ stress measurement for design of structures (mine excavations, tunnels etc.) in rock
10	Determination of shear strength of drained and undrained soil samples	2	Shear strength of soil for design of mine slopes and geo-engineering design.
11	Determination of consistency and compaction of soil	2	Soil consistency, compaction and deformation behaviour for designing any structure in soil; Soil classification and characterization

Text Books:

1. ISRM Suggested Testing Methods: E.T. Brown; and BIS codes for testing of rocks and soils
2. Handbook on Mechanical Properties of Rocks, Vol.1-4. Trans Tech. Publications: Vutukuri, V. S., R. D. Lama and S. S. Saluja,

Evaluation of Practical (weightage):

1. Regular Submission of Practical Note Book with observations (25%)
2. Attendance (25%)
3. End Semester Practical Examination: 50% (Practical Exam + Viva Voce)

Rock Mechanics and Rock Excavation Laboratories
Department of Mining Engineering,
Indian Institute of Technology (Indian School of Mines) Dhanbad

Do's and Dont's

The following instructions may be followed while conducting practical's in the above-mentioned laboratories:

1. Read the experimental manuals/procedures supplied to you in advance, carefully and completely and make your own observations on the type, procedure and care that needs to be exercised in each practical.
2. Listen to the instructions of the instructor, technical personnel and associated scholar and follow meticulously. Any doubts with regards to usage please do not hesitate to contact any one of the above.
3. Unless you are asked do not touch any equipment, samples or any part of the equipment.
4. Carry out the measurements and experiments under the supervision of the designated instructor. Do not operate or insist to operate equipment without proper permissions.
5. Do not leave any of your belongings in the laboratory and check before you leave.
6. Take a good care of yourself, the equipment and samples and leave them in good shape.
7. Any uncommon noise, smoke, vibration and leakage while operating the machine or any other nearby equipment if noticed please bring it to the notice of the concerned laboratory technical person and teacher concerned. Never venture into solve it yourself unless asked for.
8. Any impending danger foreseen please evacuate the place and inform the concerned.
9. Wear cotton clothes with loose fit and shoes before you enter the lab. Goggles as per the need shall be provided to you to wear. Wear them before you start your work.

Experiment-1

Name of the experiment: Preparation of rock samples for testing

1. SCOPE

In order to obtain valid results from tests on rock materials, careful and precise specimen preparation is required. This method outlines preparatory procedures recommended for normal rock mechanics tests.

2. COLLECTION AND STORAGE

Test material is normally collected from the field in the form of drilled cores. Field sampling procedures should be rational and systematic, and the material should be marked to indicate its original position and orientation relative to identifiable boundaries of the parent rock mass.

Good practice generally dictates that laboratory tests be made upon specimens representative of field conditions. Thus, it follows that the field moisture condition of the specimen should be preserved until the time of the test.

Ideally, samples should be moisture proofed immediately after collection either by waxing, spraying, or packing in polyethylene bags or sheet. (Example: For moisture proofing by waxing, the following procedure can be used for core that will not fail apart in handling.

- a) Wrap core in a clear thin polyethylene,
- b) Wrap in cheese cloth,
- c) Coat wrapped core with a lukewarm wax mixture to an approximate 6 mm thickness.

The wax should consist of a 1 to 1 mixture of paraffin and microcrystalline wax.

They should be transported as a fragile material and protected from excessive changes in humidity and temperature. The identification markings of all samples should be verified immediately upon their receipt at the laboratory, and an inventory of the samples received should be maintained.

Samples should be examined and tested as soon as possible after receipt; however, it is often necessary to store samples for several days or even weeks to complete a large testing program. Every care must be taken to protect stored samples against damage. Core logs of samples should be available.

3. AVOIDANCE OF CONTAMINATION

The deformation and fracture properties of rock may be influenced by air, water, and other fluids in contact with their internal (crack and pore) surfaces. If these internal surfaces are contaminated by oils or other substances, their properties may be altered appreciably and give misleading test results. Of course, a cutting fluid is required with many types of specimen preparation equipment. Clean water is the preferred fluid. Even so, one must be cognizant of the effect of moisture on the test specimens. While it may be impossible to exactly duplicate the in situ conditions even if they were known, a concerted effort should be made to simulate the environment from which the samples came. Generally, there are three conditions to be considered:

- a) Hard, dense rock and low porosity will not normally be affected by moisture. This type of material is normally allowed to air-dry prior to testing to bring all samples to an equilibrium condition. Drying at temperatures above 120°F (49°C) is not recommended as excessive heat may cause an irreversible change in rock properties.
- b) Some shales and rocks containing clay will disintegrate if allowed to dry. Usually the disintegration of diamond drill cores can be prevented by wrapping the cores as they

are drilled in a moisture proof material such as aluminum foil or chlorinated rubber, or sealing them in moisture proof containers.

- c) Mud shales and rock containing bentonites (e.g., tuff) may soften if the moisture content is too high. Most of the softer rocks can be cored or cut using compressed air to clear cuttings and to cool the bit or saw.

It is imperative to determine very early in the test program the moisture sensitivity of all types of material to be tested and to take steps to accommodate the requirements throughout the test life of the selected specimens.

4. SAMPLE SELECTION

The judicious sample selection is important for overall success of the testing programme. It follows, therefore, that test specimens should be of good quality unweathered material, should be free from incipient fractures or damage and should be of regular shape and size.

The timing of sample collection may also play an important role with regard to the end result of tests, in that collection of samples should coincide with the availability of the facilities for specimen preparation, thus avoiding long delays and storage problems which, in turn, may result in deterioration and unnecessary additional costs.

Closest teamwork of the laboratory personnel and the project engineer/geologist must be continued throughout the testing program since, as quantitative data become available, changes in the initial allocation of samples or the securing of additional samples may be necessary.

5. CORING

Virtually all laboratory coring is done with thin-wall diamond rotary bits, which may be detachable or integral to the core barrel. The usual size range for laboratory core drills is from 6-in.-diam (152.4-mm) down to 1-in. (25.4-mm) outside diameter. Typical sample diameters for uniaxial testing are 2.125 in. (54 mm). Drilling machines range from small quarry drills to modified machine shop drill presses. Almost any kind can be adapted for rock work by fitting a water swivel, but a heavy, rigid machine is desirable in order to assure consistent production of high quality core. The work block must be clamped tightly to a strong base or table so as to prevent any tilting, oscillation, or other shifting. To avoid unnecessary unclamping and rearrangement of the work block, it is desirable to have provision for traversing the drill head or the work block. Traversing devices must lock securely to eliminate any play between drill and work. The drill travel should be sufficient to permit continuous runs of at least 10 to 12 in. (254 to 304.8 mm), without need for stopping the machine. Optimum drilling speeds vary with bit size and rock type, and to some extent with condition of the bit and the characteristics of the machine. The general trend is that drill speed increases as drill diameter decreases; also, higher drill speeds are sometimes used on softer rocks. The broad range of drill speeds lies mainly between 200 and 2,000 rpm. No hard-and-fast rules can be given, but an experienced operator can easily choose a suitable speed by trial. Some core drills are hand-fed, but it is desirable to have some provision for automatic feed. The ideal feed arrangement is a constant-force hydraulic feed which can be set for each bit size and rock type, but such machines are quite rare. Constant-force feed can be improvised by means of a weight and pulley arrangement. On adapted metalworking drill presses, the automatic feed rate for a given drill size and rock type can be determined; however, since there is a danger of damaging the machine or the core barrel if too high a feed rate is used, an electrical overload breaker should be provided.

6. SAWING

For heavy sawing, a slabbing saw is adequate for most purposes. For exact sawing, a precision cutoff machine, with a diamond abrasive wheel about 10 in. (254 mm) in diameter, and a table with two-way screw traversing and provision for rotation are recommended. The speed of the wheel is usually fixed, but the feed rate of the wheel through the work can be controlled.

Clean water, either direct from house supply or re-circulated through a settling tank, is the standard cutting and cooling fluid. For crosscutting, core should be clamped in a vee-block slotted to permit passage of the wheel. By supporting the core on both sides of the cut, the problem of spalling and lip formation at the end of the cut is largely avoided. Saw cuts should be relatively smooth and perpendicular to the core axis in order to minimize the grinding or lapping needed to produce end conditions required for the various tests.



Fig.1: Rock coring and cutting machines for preparation of rock samples

7. END PREPARATION

Due to the rather large degree of flatness required on bearing surfaces for many tests, end grinding or lapping is required. Conventional surface lapping provided the most practical means of preparing flat surfaces. Procedures are essentially comparable to metal working. Quite often a special jig is constructed to hold one or more specimens in the lapping operation. For core diameters of 2-1/8 in. (54 mm) or less, a lap can be used for grinding flat end surfaces on specimens, although producing a sufficiently flat surface by this method is an art. To end-grind on the lap, a cylindrical specimen is placed in a steel carrying tube which is machined to accept core with a clearance of about 0.002 in. (0.0508 mm). At the lower end of this tube is a steel collar which rests on the lapping wheel. The method requires use of grinding compounds and, hence, is not recommended where other methods are available.

8. SPECIMEN CHECK

After the specimen has been finally ground to size, it should be submitted for inspection before testing. The checking should take account of dimensional accuracy and the conditional integrity of the specimen relative to the test concepts. In general terms, test specimens should be straight, their diameter should be constant, and the ends should be flat, parallel, and normal to the long axis. Final dimensions of the specimen are normally measured with a dial gauge, vernier

caliper, Vee block etc. and reported as listed below (sample for uniaxial compressive strength according to ISRM-1978).

- a) The diameter of the specimen should be measured to the nearest 0.1mm and its length-to-diameter ratio should be checked to have between 2.5 to 3.0.
- b) The ends of the specimen should be ground or lapped flat to within 0.02 mm.
- c) The ends of the specimen should be square to the perpendicular axis and to be within 0.001radian or 0.05mm in 50mm.
- d) The sides of the specimen should be smooth and free from any irregularities, and be straight to within 0.3mm over the entire length.
- e) Specimens should not be tested which do not meet the dimensional tolerances specified in the respective test methods.

9. REQUIREMENT OF MINIMUM NUMBER OF SPECIMENS

Test results obtained by testing individual specimens sometimes indicate a high degree of scatter and not always, representative of the present rock body. Therefore, it often necessary to conduct a greater number of test samples in order to limit costs without compromising accuracy or reliability. There are various statistical methods for obtaining minimum number of samples required for an individual test on a particular rock type. However, one of the methods considering varying degree of confidence is given below.

$$\text{Standard deviation} = S = \sqrt{\frac{\sum (x_i - x_m)^2}{N - 1}}$$

$$V = \text{coefficient of variation in the preliminary test of } N \text{ specimen} = \frac{S}{x_m} \times 100$$

The following relationship can be used for determination of number of specimens 'n' to be tested

$$n = \left[\frac{(P+1)tV}{100(P-1)} \right]^2$$

where,

$$P = \text{a number depends on permissible deviation of mean value} = \frac{100+r}{100-r}$$

r = permissible deviation of mean value or error

(i.e. for r = 10% P = 1.22)

t = depends on the confidence desired and the number of specimens tested (N) in a preliminary experimental test (See table). (i.e. for N = 4 t = 3.18 for 95 % confidence level)

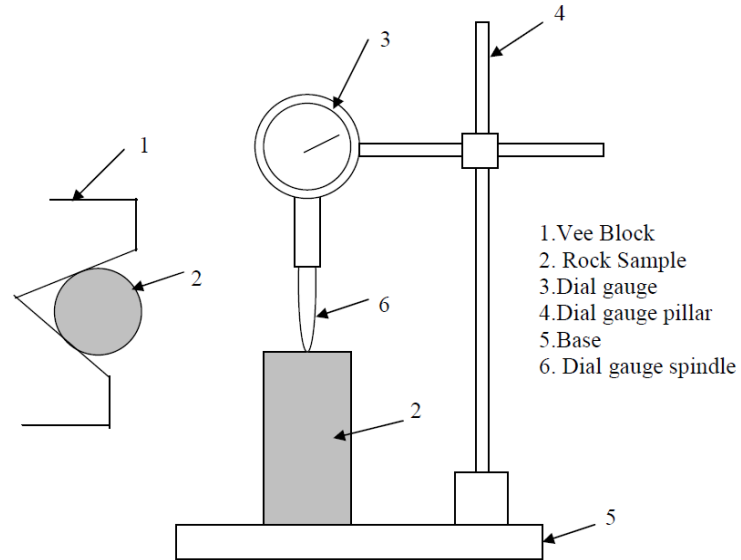


Fig.1: Typical rock sample inspection setup.

Table: Some critical value of 't'
t

N	DF*	t			Confidence level %
		90	95	99	
3	2	2.92	4.30	9.92	
4	3	2.35	3.18	5.84	
5	4	2.13	2.78	4.60	
6	5	2.02	2.57	4.03	
7	6	1.94	2.45	3.71	
8	7	1.90	2.36	3.50	
9	8	1.86	2.31	3.36	
10	9	1.83	2.26	3.25	
11	10	1.81	2.23	3.17	
12	11	1.80	2.20	3.11	

* Degrees of freedom

References:

International Society for Rock Mechanics Suggested Methods "Rock Characterization, Testing and Monitoring" Editor E.T. Brown, Pergamon Press 1981.

Experiment-2

Name of the experiment: Determination of uniaxial compressive strength of rock samples

INTRODUCTION

One of the important engineering properties of the rock, which plays an essential role in designing various geotechnical applications such as underground mines, tunnel and dam is Uniaxial Compressive Strength (UCS) of the core specimen. Compressive strength is a key value for design of underground structures. The determination of compressive strength involves the use of a cylindrical specimen of rock loaded axially between platens in a testing machine. The stress value at failure is defined as the compressive strength of the specimen. There is no load along an axis perpendicular to the loading axis. The loading condition simulates the condition of pillars supporting the mine roof in case of underground mines. The Uniaxial compressive strength depends on friction between platens and end surfaces, specimen geometry, rate of loading and water content. If the friction between loading plate and the rock specimen is more, the specimen will fail in shear, whereas in case of a smooth contact between the loading plate and the specimen, the specimen will fail in tension. The compressive strength of rocks usually increases with increase in the rate of loading of specimen. The strength of the specimen usually decreases with increase in their size and when the ratio of length to diameter increases the compressive strength will be decreases. Dry specimens have high compressive strength and saturated specimens have low values.

SCOPE

This method of test is intended to measure the uniaxial compressive strength of a rock sample in the form of specimens of regular geometry. The test is mainly intended for strength classification and characterization of intact rock.

APPARATUS

(a) A suitable machine (Figure 1) shall be used for applying and measuring axial load to the specimen. It shall be of sufficient capacity and capable of applying load at a rate conforming to the requirements.



Fig. 1: Universal Testing Machine

- (b) Equipments for measuring dimensions of the rock sample as shown in Figure 2.
(c) Equipments for measuring weight of the rock sample as shown in Figure 2.



Fig. 2: Digital Vernier and Electronic balance

PROCEDURE

- (a) Test specimens shall be right circular cylinders having a height to diameter ratio of 2.5-3.0 and a diameter preferably of not less than NX core size, approximately 54 mm. The diameter of the specimen should be related to the size of the largest grain in the rock by the ratio of at least 10:1.
- (b) The ends of the specimen shall be flat to 0.02 mm and shall not depart from perpendicularity to the axis of the specimen by more than 0.001 radian (about 3.5 min) or 0.05 mm in 50 mm.
- (c) The sides of the specimen shall be smooth and free of abrupt irregularities and straight to within 0.3 mm over the full length of the specimen.
- (d) The use of capping materials or end surface treatments other than machining is not permitted.
- (e) The diameter of the test specimen shall be measured to the nearest 0.1 mm by averaging three diameters measured at right angles to each other at about the upper-height, the mid-height and the lower height of the specimen. The average diameter shall be used for calculating the cross-sectional area. The height of the specimen shall be determined to the nearest 1.0 mm.
- (f) Samples shall be stored, for no longer than 30 days, in such a way as to preserve the natural water content, as far as possible, and tested in that condition.
- (g) Load on the specimen shall be applied continuously at a constant stress rate such that failure will occur within 5-10min of loading, alternatively the stress rate shall be within the limits of 0.5-1.0 MPa/sec.
- (j) The number of specimens tested should be determined from practical considerations but at least five are preferred.

The uniaxial compressive strength (σ_c) of the specimen shall be calculated by dividing the maximum load (P) carried by the specimen during the test, by the original cross-sectional area (A).

$$\sigma_c = \frac{P}{A}$$

Observations:

Sample No.	Rock Type	Diameter			Avg. Diameter (mm)	Length			Avg. Length (mm)	Weight (kg)	Breaking load (tonne)	Total time taken (sec.)	Type of failure
		Top (mm)	Middle (mm)	Bottom (mm)		Top (mm)	Middle (mm)	Bottom (mm)					
<u>Example</u>	Charnockite	54.28	54.28	54.32	54.29	138.3	138.32	138.42	138.35	0.871	28.35	115	Shear
1.													
2.													

Calculations:

Calculate the cross-sectional area of the specimen as follows:

$$A = (\pi D^2) / 4$$

where: A = cross-sectional area, in (mm²)

D = average specimen diameter, in (m)

Calculate the volume of the specimen as follows:

$$V = A \times L$$

where: V = volume, in (m³)

A = cross-sectional area, in (mm²)

L = specimen length, in (m)

Calculate the specimen unit weight as follows:

$$UW = (M \times g) / V$$

where: UW = specimen unit weight, in (N/ m³)

M = specimen mass, in (kg)

W = specimen weight, in (N)

V = volume, (m³)

Calculate the compressive strength in the test specimen from the maximum compressive load on the specimen and the initial computed cross-sectional area as follows:

$$\sigma_c = \frac{P}{A}$$

where: σ_c = compressive strength, in MPa

P = maximum load, in (kN)

A = cross sectional area, in m²



Before



After

Examples of Calculations:

Given:

Average diameter = 54.29mm

Length = 138.35mm.

Mass = 0.871kgs

Maximum load = 28.35 tonnes

Calculation:

$$L/D = 138.35 / 54.29 = 2.55$$

$$A = [\pi \times (54.29)^2] / 4 = 23.15 \times 10^{-4} \text{ m}^2$$

$$V = (23.15 \times 10^{-4}) \times (0.13835) = 3.202 \times 10^{-4} \text{ m}^3$$

$$\text{Density } (\rho) = 0.871 / 3.202 \times 10^{-4} = 2720 \text{ kg/ m}^3$$

$$\text{Unit weight } (\gamma) = \rho \times g = 2720 \times 9.81 = 26683 \text{ N/ m}^3$$

$$\sigma_c = \frac{\text{Breaking load} \times g}{A} = \frac{28.35 \times 10^3 \times 9.81}{23.13 \times 10^{-4}} = \underline{120.08 \text{ MPa}}$$

$$\text{Rate of stress} = (120.08 \text{ MPa}) / 115 = \underline{1.04 \text{ MPa/Sec.}}$$

References:

International Society for Rock Mechanics Suggested Methods "Rock Characterization, Testing and Monitoring" Editor E.T. Brown, Pergamon Press 1981.

Experiment-3

Name of the experiment: Determination of tensile strength of rock samples by Brazilian method

INTRODUCTION

Tensile strength may be defined as the maximum stresses developed in a specimen in a tension test performed to rupture it. In rock mechanics knowledge of tensile strength of rock is important in designing roofs and domes of underground openings. A rock slab or beam subjected to bending also experiences a tensile stress. Although rock is weak in tension, an average tensile strength of rock has been found to be one tenth of its compressive strength. Although the principle of measurement of tensile strength is to apply an axial tensile force to a rock specimen the method is not successful due to difficulties in making specimen of rock samples for the test. With rock, it is very difficult to make specimen of such a shape. An alternative is to grip the cylindrical sample at two ends with some fixing material, and then to apply the tensile force at the two ends. Due to the gripping difficulties, the tensile strength of rock is estimated by indirect method (Brazilian test).

SCOPE

This test is intended to measure the uniaxial tensile strength of prepared rock specimens indirectly by the Brazil test. The justification for the test is based on the experimental fact that most rocks in biaxial stress fields fail in tension at their uniaxial tensile strength when one principal stress is tensile and the other finite principal stress is compressive with a magnitude not exceeding three times that of the tensile principal stress.

APPARATUS

- (a) Two steel loading jaws designed so as to contact a disc-shaped rock sample at diametrically-opposed surfaces over an arc of contact of approx. 10° at failure. The suggested apparatus to achieve this is illustrated in Figure 1. The critical dimensions of the apparatus are the radius of curvature of the jaws, the clearance and length of the guide pins coupling the two curved jaws and the width of the jaws. These are as follows: Radius of jaws - $1.5 \times$ specimen radius; guide pin clearance - permit rotation of one jaw relative to the other by 4×10^{-3} rad out of plane of the apparatus (25 mm penetration of guide pin with 0.1 mm clearance); width of jaws - $1.1 \times$ specimen thickness. The remaining dimensions can be scaled from Figure 1. The upper jaw contains a spherical seating conveniently formed by a 25 mm diameter half-ball bearing.
- (b) Double thickness (0.2 - 0.4mm) adhesive paper strip (masking tape) with a width equal to or slightly greater than the specimen thickness.
- (c) A suitable machine for applying and measuring compressive loads to the specimen. It shall be of sufficient capacity and be capable of applying load at a rate conforming to the requirements set out in section.
- (d) Equipments for measuring dimensions of the rock sample

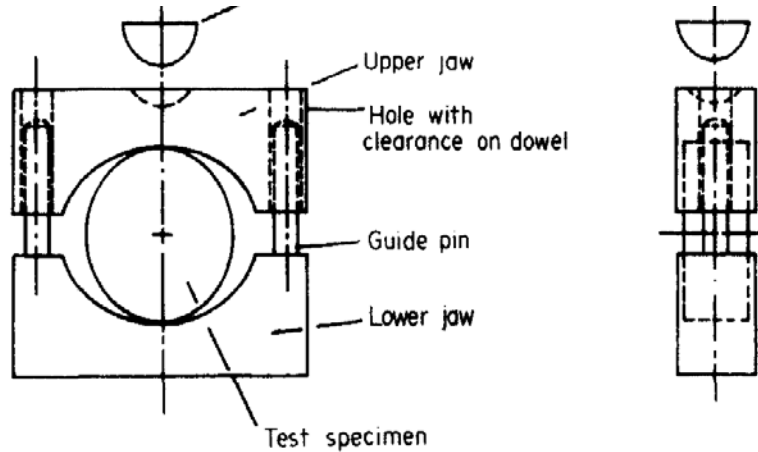


Fig. 1: Apparatus for Brazilian test

PROCEDURE

- The test specimens should be cut and prepared using clean water. The cylindrical surfaces should be free from obvious tool marks and any irregularities across the thickness of the specimen should not exceed 0.025 mm. End faces shall be flat to within 0.25 mm and square and parallel to within 0.25°.
- The specimen diameter shall not be less than NX core size, approximately 54mm, and the thickness should be approximately equal to the specimen radius.
- The test specimen shall be wrapped around its periphery with one layer of the masking tape and mounted squarely in the test apparatus such that the curved platens load the specimen diametrically with the axes of rotation for specimen and apparatus coincident.
- Load on the specimen shall be applied continuously at a constant rate such that failure in the weakest rocks occurs within 15-30 s. A loading rate of 200 N/s is recommended (Figure 2).
- The number of specimens per sample tested should be determined from practical considerations, but normally ten are recommended.

CALCULATIONS

The tensile strength of the specimen σ_t , shall be calculated by the following formula:

$$\sigma_t = 0.636 P/Dt \text{ (MPa)}$$

where P is the load at failure(N), D is the diameter of the test specimen (in mm), t is the thickness of the test specimen measured at the center (in mm).

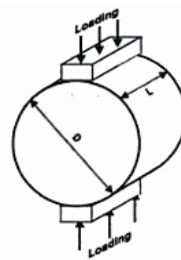


Fig. 2: Brazilian testing arrangement for determining indirect tensile strength

Tensile strength test is to verify tensile strength of rock samples or its resistance against fracturing. Direct tensile test on rock sample is relatively difficult to undertake and Brazilian test offers an indirect method to measure tensile strength. The term ‘indirect tensile’ implies that load is applied under compression. Figure 3 shows typical mode of failure associated with tensile stress.



Figure 3: Mode of failure of rock samples in Brazilian test

Observations:

Sample No.	Rock Type	Diameter	Avg. Diameter	Thickness	Avg. Thickness	Weight	Breaking load	Total time taken	Circumferential area	Stress rate or Rate of	Density	Avg. Density	Type of failure	Tensile strength	Avg. Tensile strength

References:

International Society for Rock Mechanics Suggested Methods "Rock Characterization, Testing and Monitoring" Editor E.T. Brown, Pergamon Press 1981.

Experiment-4

Name of the experiment: Determination of modulus of elasticity and poisson's ratio using strain gauges

SCOPE

This method of test is intended to determine stress-strain curves and Young's modulus and Poisson's ratio in uniaxial compression of a rock specimen of regular geometry. The test is mainly intended for classification and characterization of intact rock.

APPARATUS

- (a) A suitable machine shall be used for applying and measuring axial load to the specimen. It shall be of sufficient capacity and capable of applying load at a rate conforming to the requirements.
- (b) Electrical resistance strain gauges for measuring strain of the rock sample. Their design shall be such that the average of two circumferential and two axial strain measurements, equally spaced, can be determined for each increment of load. The length of the gauges over which axial and circumferential strains are determined shall be at least ten grain diameters in magnitude and the gauges should not encroach within $D/2$ of the specimen ends, where d is the diameter of the specimen.
- (c) An apparatus for recording the loads and deformations; preferably an X-Y recorder capable of direct plotting of load-deformation curves.

PROCEDURE

- (a) Test specimens shall be right circular cylinders having a height to diameter ratio of 2.5-3.0 and a diameter preferably of not less than NX core size, approximately 54 mm. The diameter of the specimen should be related to the size of the largest grain in the rock by the ratio of at least 10:1.
- (b) The ends of the specimen shall be flat to 0.02 mm and shall not depart from perpendicularity to the axis of the specimen by more than 0.001 radian (about 3.5 min) or 0.05 mm in 50 mm.
- (c) The sides of the specimen shall be smooth and free of abrupt irregularities and straight to within 0.3 mm over the full length of the specimen.
- (d) The use of capping materials or end surface treatments other than machining is not permitted.
- (e) The diameter of the test specimen shall be measured to the nearest 0.1 mm by averaging three diameters measured at right angles to each other at about the upper-height, the mid-height and the lower height of the specimen. The average diameter shall be used for calculating the cross-sectional area. The height of the specimen shall be determined to the nearest 1.0 mm.
- (f) Moisture can have a significant effect on the deformability of the test specimen. When possible, in situ moisture conditions should be preserved until the time of the test. When the characteristic of the rock material under conditions varying from saturation to dry is required, proper note shall be made of moisture conditions so that correlation between deformability and moisture content can be made. Excess moisture can create a problem of adhesion of strain gauges which may require making a change in moisture content of the sample.
- (g) Load on the specimen shall be applied continuously at a constant stress rate such that failure will occur within 5-10 min of loading, alternatively the stress rate shall be within the limits of 0.5-1.0 MPa/sec.

(h) Load and axial and circumferential strains or deformations shall be recorded at evenly spaced load intervals during the test. At least ten readings should be taken over the load range to define the axial and diametric stress-strain curves.

(j) The number of specimens tested should be determined from practical considerations but at least five are preferred.

OBSERVATIONS

Sl. No.	Rock type	Axial load	Axial strain	Diametric strain
1.				

CALCULATIONS

(a) Axial strain, ε_a , and diametric strain, ε_d , may be recorded directly from strain indicating equipment or may be calculated from deformation readings depending upon the type of instrumentation. Axial strain is calculated from the equation

$$\varepsilon_a = \frac{\Delta l}{l_o}$$

where, l_o = original measured axial length

Δl = change in measured axial length (defined to be positive for a decrease in length)

(b) Diametric strain may be determined either by measuring the changes in specimen diameter or by measuring the circumferential strain. In the case of measuring the changes in diameter, the diametric strain is calculated from the equation

$$\varepsilon_d = \frac{\Delta d}{d_o}$$

where, d_o = original undeformed diameter of the specimen

Δd = change in diameter (defined to be negative for an increase in diameter)

In the case of measuring the circumferential strain ε_c , the circumference is $C = \pi d$, thus the change in circumference is $\Delta C = \pi \Delta d$. Consequently, the circumferential strain, ε_c , is related to diametric strain, ε_d , by

$$\varepsilon_c = \frac{\Delta C}{C_o} = \frac{\Delta d}{d_o}; \quad \text{So that } \varepsilon_c = \varepsilon_d$$

where, C_o = and d_o are original specimen circumference and diameter, respectively.

(c) The compressive stress in the test specimen, σ , is calculated by dividing the compressive load P on the specimen by the initial cross-sectional area, A_o .

Thus

$$\sigma = \frac{P}{A_o}$$

where in this test procedure, compressive stresses and strains are considered positive.

(d) Figure 1 illustrates typical plot of axial stress versus axial and diametric strains. These curves show typical behaviour of rock materials from zero stress up to ultimate strength, σ_u . The complete curves give the best description of the deformation behaviour at low and high stress levels.

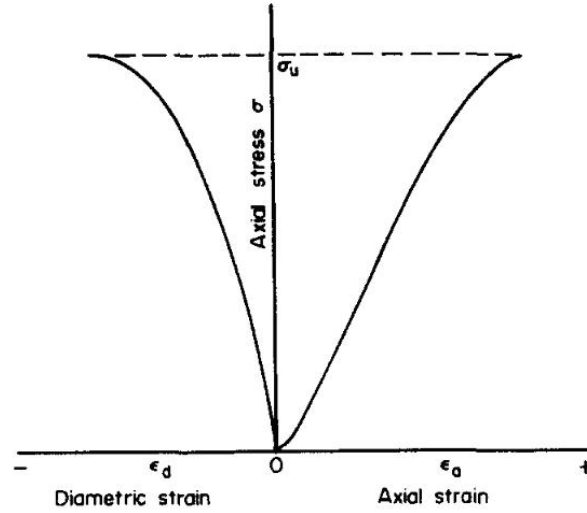


Fig. 1: Format for graphical presentation of axial and diametric stress-strain curves

(e) Axial young's modulus, E (defined as the ratio of the axial stress change to axial strain produced by the stress change) of the specimen may be calculated using any one of several methods employed in accepted engineering practice. The most common methods, listed in Figure 2, are as follows:

(i) Tangent Young's modulus, E_t , is measured at a stress level which is some fixed percentage of the ultimate strength (Figure 2a). It is generally taken at a stress level equal to 50% of the ultimate uniaxial compressive strength.

(ii) Average Young's modulus, E_{av} , is determined from the average slopes of the more or less straight line portion of the axial stress-axial strain curve (Figure 2b).

(iii) Secant Young's modulus, E_s , is usually measured from zero stress to some fixed percentage of the ultimate strength (Figure 2c), generally at 50%.

(f) Poisson's ratio, ν , may be calculated from the equation

$$\nu = - \frac{\text{slope of axial stress-strain curve}}{\text{slope of diametric stress-strain curve}}$$

$$\nu = - \frac{E}{\text{slope of diametric stress-strain curve}}$$

Poisson's ratio in this equation has a positive value, since the slope of the diametric curve is negative by the conventions used in this procedure.

(g) The volumetric strain, ϵ_v , for a given stress level is calculated from the equation

$$\epsilon_v = \epsilon_a + 2\epsilon_d$$

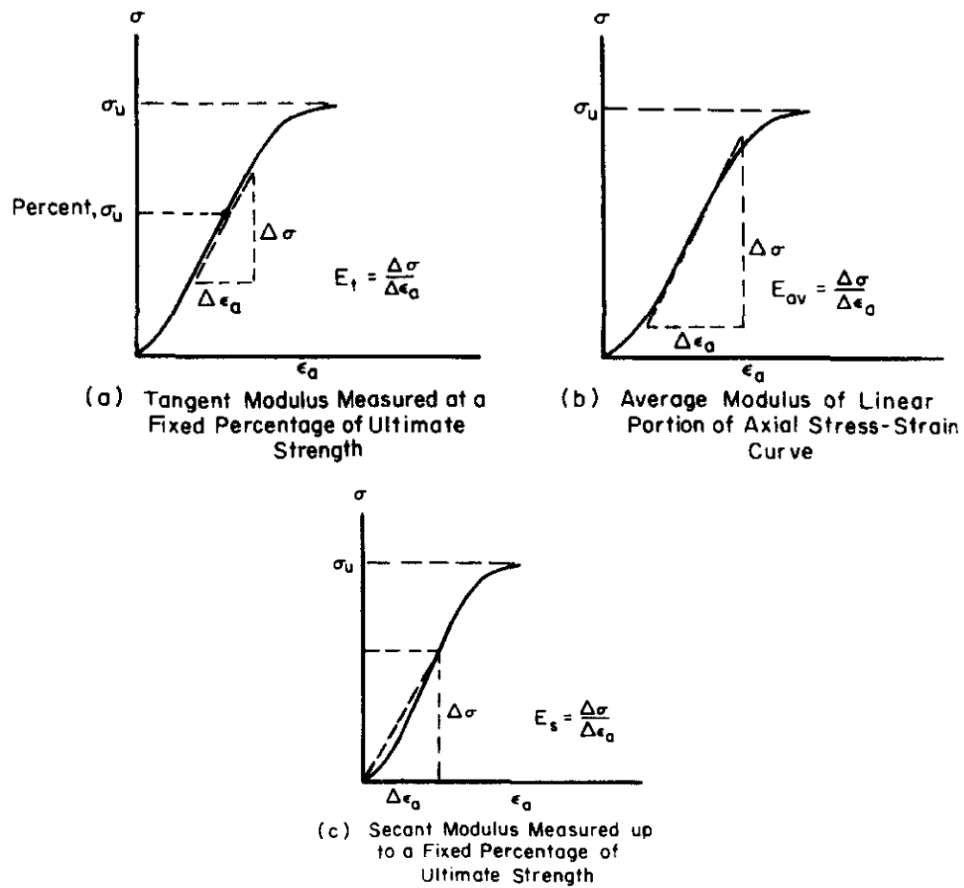


Fig. 2: Methods for calculating Young's modulus from axial stress-strain curve

References:

International Society for Rock Mechanics Suggested Methods "Rock Characterization, Testing and Monitoring" Editor E.T. Brown, Pergamon Press 1981.

Experiment-5

Name of the experiment: Determination of point load strength index of rock samples

INTRODUCTION

Point load testing is used to determine rock strength indexes in geotechnical practice. This index can be used to estimate other rock strength parameters. In a Point Load Test, a sample of rock is mounted between two pointed platens and pressure is applied until failure of the sample occurs. The peak applied load is recorded and used to calculate the Point Load Index. The samples are tested to know minimum and maximum point load strength for estimating anisotropic index.

SCOPE

- a) The Point Load Strength test is intended as an index test for the strength classification of rock materials. It may also be used to predict other strength parameters with which it is correlated for example uniaxial tensile and compressive strength.
- b) The test measures the Point Load Strength Index ($I_{s(50)}$) of rock specimens, and their Strength Anisotropy Index ($I_{a(50)}$) which is the ratio of Point Load Strengths in directions which give the greatest and least values.
- c) Rock specimens in the form of either core (the diametral and axial tests), cut blocks (the block test), or irregular lumps (the irregular lump test) are broken by application of concentrated load through a conical platens, which may be tested without any treatment.
- d) The testing may be carried out either in the laboratory or in field at the drilling site.
- e) These tests are not reliable if point load strength index (I_s) is less than 1 MPa.

PURPOSE

It is determined for mainly classify rock and determining uniaxial compressive strength of rock.

APPARATUS

Point load strength apparatus (Figure 1) is consisting of (1) Loading Frame (2) Hydraulic Jack and Accessories (3) Pressure Gauges (4) Distance Measuring System

(1) Loading Frame

The frame shall be sufficiently rigid and shall be so designed that the load application is in plumb. The distance between the top and the bottom bearing plates of this frame shall accommodate a hydraulic jack, two loading platens and platen to platen clearance for testing of rock specimens in the range of 15-100 mm. This frame shall have adequate adjustment to align perfectly the loading axis passing through the centre of the bearing plates and loading platens at all the positions of the ram of the hydraulic jack. Loading platens shall be conical with 60° angle and the radius of curvature of the cone truncation shall be 5 mm as shown in Figure 2. The platens should be of hard materials such as tungsten carbide or hardened steel so that they remain undamaged during frequent testing.

(2) Hydraulic Jack and Accessories

The loading capacity of hydraulic jack should be of sufficient capacity to break the largest and strongest specimen to be tested. There shall not be any play between the ram and the jack. The ram shall have low friction seals. The friction between the ram and the jack shall be less than 50 N (5 kgf). The system is to be resistant to hydraulic shock and vibration so that accuracy of

readings is not adversely affected by repeated testing. If quick-retracting ram is used to reduce the delay between tests, either the ram return spring force and ram friction should together be less than about 5 percent of the smallest load to be measured during testing, or an independent load cell rather than an oil pressure gauge should be used for load determination. This force can be significant when testing weaker and smaller specimens.

(3) Pressure Gauges

Bourden type pressure gauges of 200 mm diameter shall be equipped with maximum reading indicator. The gauges shall be fitted with a hydraulic 'snubber' of orifice type to protect the gauges against sudden decompression. These gauges shall have the following ranges:

- a) Gauge No. 1 - Maximum loading capacity of 25 kN (2500 kgf) to read up to 0.25 kN (25 kgf).
- b) Gauge No. 2 -Maximum loading capacity of 100 kN (10000 kgf) to read up to 0.50 kN (50 kgf).

(4) Distance Measuring System

The distance measuring system, a direct reading scale is fitted to permit measurement of the distance 'D' between specimen platen contact points. Measurements of 'D' should be to an accuracy of ± 2 percent of D or better irrespective of the size of specimen tested.



Fig.1: Point Load Apparatus

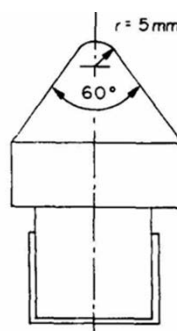


Fig. 2: Platen shape and tip radius

SAMPLE

The required size of rock samples are shown in the Figure 3 for various tests.

PROCEDURE

- The equivalent diameter D shall be measured in mm.
- The specimen core or fragment should be tested at the water content as obtained in the field and/or after soaking them for 7 days depending on requirement.
- The specimen shall be held horizontal between the two loading platen.
- The correct position of the specimen shall be checked first by giving longitudinal rotation to see the distance between loading points is minimum.

The Diametral Test:

- Core specimens with length/diameter ratio greater than 1.5 are suitable for diametral testing.
- There should preferably be at least 10 test specimens per sample and even more, if the sample is heterogeneous or anisotropic.
- The specimen is inserted in the test machine and the platens closed to make contact along a core diameter, ensuring the distance L between contact point and the nearest free end is at least 0.75 times the core diameter (Figure 3a).
- The load shall then be applied to the core specimen such that failure occurs within 10-60 sec. and the failure load P is recorded. The test shall be rejected as invalid if the fracture surface passes through only one loading point (Figure 4d).

Axial Test:

- Core specimens with length/diameter ratio of 0.3 to 1.0 are suitable for axial testing (Figure 3b). Long pieces of core can be tested diametrically to produce suitable length for subsequent axial testing; alternatively, suitable specimen can be obtained by saw cutting or chisel splitting.
- There should preferably be at least 10 test specimens per sample and even more, if the sample is heterogeneous or anisotropic.
- The specimen is inserted in the test machine and the platens closed to make contact along a line perpendicular to the core end faces.
- The distance D between platen contact points is recorded to $\pm 2\%$. The specimen width W perpendicular to the loading direction is recorded to $\pm 5\%$. In axial test W is equal to the diameter of axial core.
- The load shall then be applied to the specimen such that failure occurs within 10-60 sec and the failure load ' P ' is recorded. The test should be rejected as invalid if the fracture surface passes through only one loading point (Figure 4e).

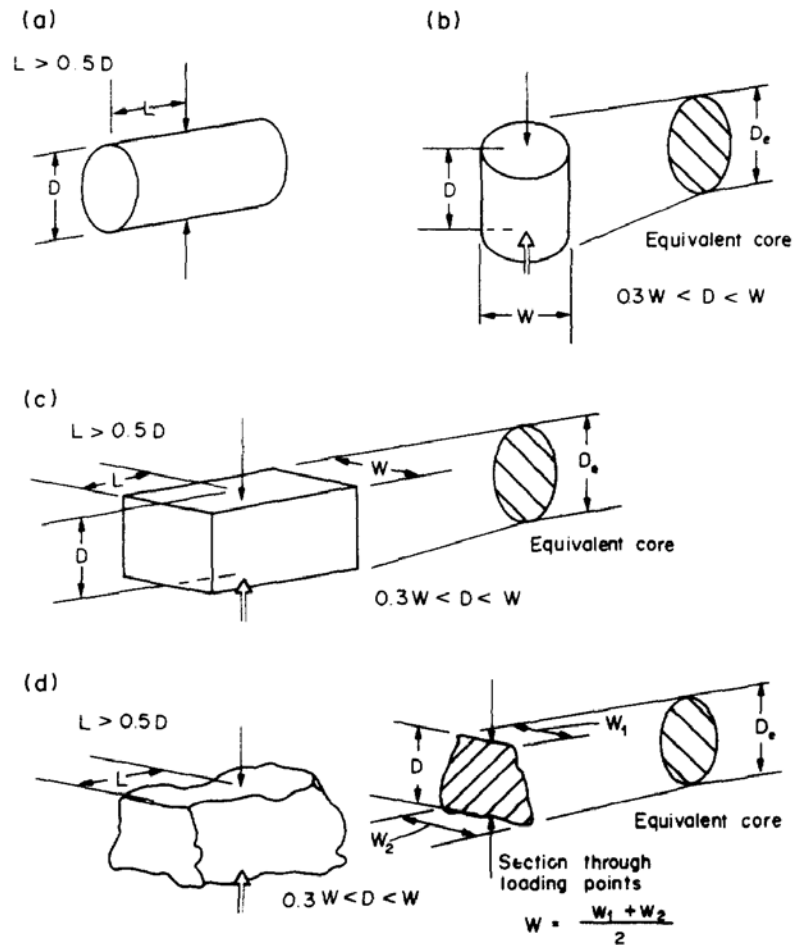


Fig. 3: Rock sample size for (a) Diametral test (b) Axial test (c) Block test, and (d) Irregular lump test

The Block and Irregular Lump Tests:

- Rock blocks or lumps of size 50 ± 35 mm and of shape shown in Figure 3(c) and (d) are suitable for the block and irregular lump tests. The ratio of D/W should be between 0.3 and 1.0, preferably close to 1.0. The distance L (Figure 3c and d) should be at least $0.5 D$.
- There should preferably be at least 10 test specimens per sample and even more, if the sample is heterogeneous or anisotropic.
- The specimen is inserted in the test machine and the platens closed to make contact with the smallest dimensions of the block or lump, away from the edges and corners (Figure 3c and d).
- The distance D between platen contact points is recorded to ± 2 %. The smallest specimen width W perpendicular to the loading direction is recorded to ± 5 %. If the sides are not parallel, then W is obtained (Figure 3d) from W_1 and W_2 and, calculated as follows:

$$W = \frac{W_1 + W_2}{2}$$

- The load shall then be applied to the specimen such that failure occurs within 10-60 sec and the failure load P is recorded. The test should be rejected as invalid if the fracture surface passes through only one loading point (Figure 4d and e).

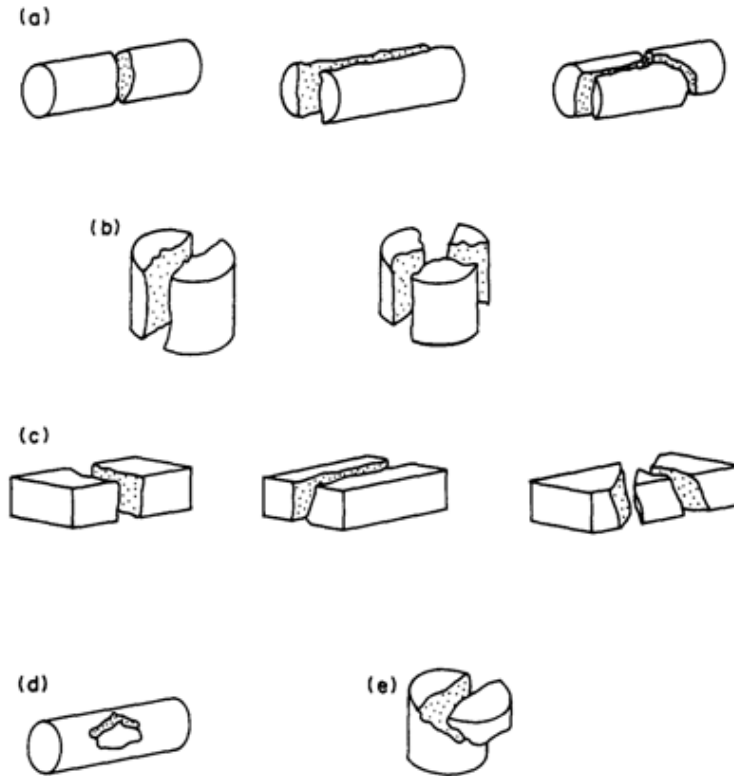


Fig. 4: Typical modes of failure for valid and invalid tests. (a) Valid diametral tests; (b) valid axial tests; valid block tests; (d) invalid core test; (e) invalid axial test

Observations:

Sample No.	Rock Type	Length of Specimen			Avg. Length of Specimen	Diameter of Specimen			Avg. Diameter of Specimen	Initial reading of scale	Final reading of scale	Load at failure
		L ₁	L ₂	L ₃		D ₁	D ₂	D ₃				

CALCULATION AND SIZE CORRECTION

The uncorrected Point Load Strength Index (I_s) is calculated from the following equation

$$I_s = \frac{P}{D_e^2}$$

where, D_e is the equivalent core diameter, is given by:

$$D_e^2 = D^2 \text{ for diametral tests;}$$

$$= 4A/\pi \text{ for axial, block and lump tests;}$$

and

$$A = WD = \text{minimum cross-sectional area of a plane through the platen contact points.}$$

The corrected Point Load Strength Index ($I_{s(50)}$) is

$$I_{s(50)} = F \times I_s \text{ and } F = \left(\frac{D_e}{50} \right)^{0.45}$$

where

I_s = point load strength index (for the core size other than 50 mm diameter) in MN/m^2 ,

$I_{s(50)}$ = point load strength index [for the Standard (core diameter = 50 mm) core size] in MN/m^2 , (kgf/cm^2);

P = failure load in N (kgf);

D_e = equivalent core diameter in mm; and

F = Correction factor

Uniaxial Compressive Strength

The uniaxial compressive strength of rock may be predicted from the following correlation:

$$\sigma_c = 22 \times I_{s(50)}$$

where

σ_c = uniaxial compressive strength in MN/m^2 (kgf/cm^2), and

$I_{s(50)}$ = corrected point load strength.

References:

International Society for Rock Mechanics Suggested Methods "Rock Characterization, Testing and Monitoring" Editor E.T. Brown, Pergamon Press 1981.

Experiment-6

Name of the experiment: Determination of Protodyakonov strength index of rock samples

INTRODUCTION

Impact strength index was first developed by Protodyakonov (1926) by M.M. Protodyakonov senior. He proposed to evaluate the mechanical properties of rocks by means of an index known as Protodyakonov strength index. Since then this index has found wide application in mining. The index gives the resistance of rock impact or dynamic loading to destruction of rock.

This index being used for various purposes such as:

- a) Classification of rock and rock mass
- b) Estimation of uniaxial axial compressive strength of rock in laboratory and in situ
- c) Assessing drillability and cuttability of mine rock

The following empirical formula has been derived for the evaluation of the drillability of pneumatic percussive blast hole drill with the help of Protodyakonov strength index (f) of rock.

$$t = \frac{d^2 (f + 8.6)}{3000N} \quad (\text{min/ } m)$$

where, t = drilling time of blast hole, min/m

d = blast hole diameter, mm

N = capacity of pneumatic drill, kwt

PROTODYAKONOV APPARATUS

Protodyakonov found that pounding of rock produces dust proportionate to the compressive strength or hardness rock. The volume of dust produced is expressed in form of an index known as Protodyakonov strength index. Based on the above principle he designed the Protodyakonov strength index test where rock samples are subjected to pounding or dynamic load by dropping a mass for a fixed height and given number of times. The fine dust produced to pounding action is measured to strength index. It is consisting of two units which are made up of cylindrical steel tubes and bars as shown in figure: (1) Pounding Apparatus (2) Volumometer

Specimen requirements

Five test rock samples should be taken. The volume of each rock sample should be around 20 cm³ consisting of number of lumps of 20-40mm size.

TEST PROCEDURE

Each sample should be placed in the pounding apparatus and pounded with a 2.4 kg drop weight falling from the height of 600mm. The number of pounding, n , of the drop weight on each sample varies from 5 to 15. The stronger rock sample, the greater the number of impact should be. All five test sample of pounded rock should be placed in the sieve of 0.5mm and sieved together. Dust, which pass through the 0.5mm sieve, should be pored into the tube of volumometer and the height of the column of dust in the volumometer should be recorded in millimeter.

CALCULATION

Protodyakonov strength index is determined by

$$f = \frac{20n}{h}$$

where, f = Protodyakonov strength index

n = number of pounding (5-15)

h = height of dust measured in volumometer in mm

Protodyakonov strength index being used for estimating uniaxial axial compressive strength of rock (UCS) from the following empirical relationship:

$$\text{UCS} = 100 \times f \quad (\text{kg/cm}^2)$$

Protodyakonov strength index is found to be directly proportional to the specific work rock destruction (A) under uniaxial axial compression and related to the volume of fine dust obtained as given below.

$$A \cong 0.53 f \quad \text{kg / cm}^3$$

As known:

$$A = \frac{\sigma_c^2}{2E}$$

where σ_c = uniaxial axial compressive strength of rock sample (kg/cm^2)

E = Young's modulus of elasticity of rock (kg/cm^2)

Hence, by substituting value of E and f uniaxial axial compressive strength of rock is calculated.

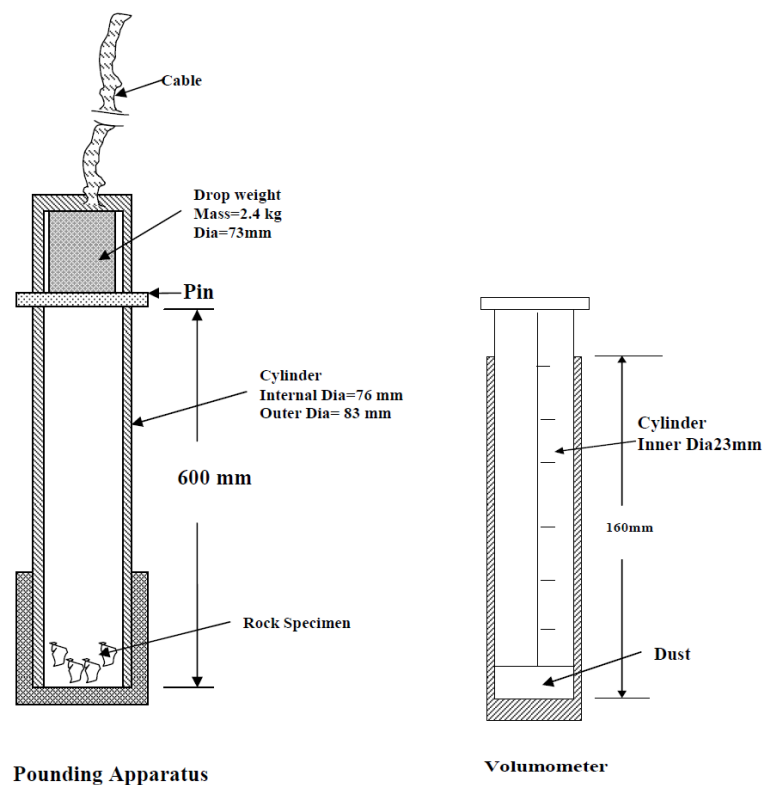


Fig.1: Protodyakonov Apparatus

Protodyakonov strength index along with the Shore scleroscope hardness index has been used for estimating UCS from the following empirical relationship.

$$\sigma_c \cong 1050 \sqrt{\frac{f}{(154/h)-1}} \quad (kg/cm^2)$$

where, h = Shore scleroscope hardness index of rock



Fig.2: Protodyakonov apparatus for determination of Protodyakonov strength index and compressive strength

Experiment-7

Name of the experiment: Determination of slake durability index of rock

1. SCOPE

1.1 This test is intended to assess the resistance offered by a rock sample to weakening and disintegration when subjected to two standard cycles of drying and wetting in a slaking fluid usually water.

2. TERMINOLOGY

2.0 For the purpose of this standard, the following definition shall apply.

2.1 Slaking Fluid - The fluid used in slake durability test is termed as slaking fluid. Tap water at 20°C, approximating to normal ambient temperature is specified for the test.

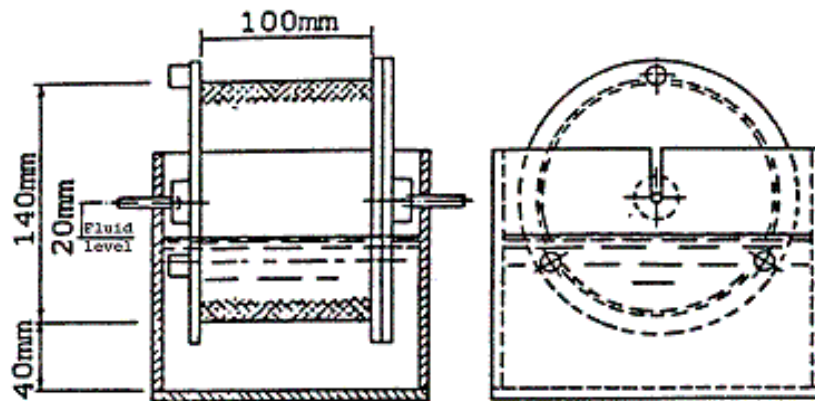


Fig. 1: A schematic view of Slake durability Apparatus



Fig: Slake durability Apparatus

3.1 The test drum shall be made from a 2.0 mm wire mesh of unobstructed length of 100.0 mm and diameter 140 mm with a solid fixed base.

The drum shall be able to withstand a temperature of 105°C. The drum shall have solid removable lid. The drum shall be sufficiently strong to retain its shape during use, but neither the exterior of the mesh nor the interior of the drum should be obstructed, by reinforcing members.

3.2 A trough, to contain the test drum supported with its axis horizontal in a manner allowing rotation, capable of being filled with a slaking fluid, such as water, to a level

20 mm below the drum axis. The drum shall be mounted to allow 40 mm unobstructed clearance between the trough and the base of the mesh.

3.3 A motor drive, capable of rotating the drum at a speed of 20 rev/ min, the speed to be held constant to within 5 percent for a period of 10 minutes.

3.4 An oven, capable of maintaining a temperature of $105 \pm 5^{\circ}\text{C}$ for a period of at least 12 hours.

3.5 A balance, capable of weighing the drum plus sample to an accuracy of 0.5 g.

4. TEST PROCEDURE

4.1 A representative sample comprising ten rock lumps should be selected, each weighing 40-60 g to give a total sample weight of 400-600 g. Lumps should be roughly spherical in shape and corners should be rounded off during preparation.

4.2 The sample shall be placed in clean drum and dried to constant weight at a temperature of $105 \pm 5^{\circ}\text{C}$. The weight A of the drum plus sample shall be recorded. The sample should then immediately be used for testing purposes.

4.3 The lid shall be then placed, the drum mounted in the trough and coupled to the motor.

4.4 The trough shall be filled with slaking fluid, usually tap water at 20°C , to a level of 20 mm below the drum axis, and the drum rotated at 20 rev/min for a period of 10 minutes.

4.5 The drum shall be removed from the trough and the lid removed from the drum. The drum plus retained portion of sample shall be dried to constant weight at $105 \pm 5^{\circ}\text{C}$. The weight B of the drum plus retained portion of the sample shall be recorded.

4.6 The test shall be repeated as in 4.4 and 4.5 for a further period of 10 minutes. The weight C of the drum plus retained portion of the sample shall be recorded.

4.7 The drum shall then be brushed clean and its weight D recorded

5. CALCULATION

5.1 The slake-durability index (second cycle) is calculated as percentage ratio of final to initial dry sample weights as follows:

$$\text{Slake durability Index (Id}_2\text{)} = \frac{C-D}{A-D} \times 100$$

6. REPORTING OF RESULTS

6.1 The report shall include the following information for each sample tested:

- a) The slake-durability index (second cycle) to nearest 0.1 percent; and
- b) The nature and temperature of the slaking fluid, usually tap water at 20°C .

6.1.1 If required additional information for details of fragments retained in the drum and material passing through the drum may also be given.

6.2 A tentative sub-division of the slake-durability scale may be used for classification as given in Table 1.

Table 1: Classification of slake durability index

Slake-Durability Id (Percentage)	Classification
0-25	Very Low
Over 25-50	Low
Over 50-75	Medium
Over 75-90	High
Over 90-95	Very high
Over 95-100	Extremely high

Experiment-8

Name of the experiment: Determination of triaxial strength of rock samples – Individual test

SCOPE OF THE TEST

The objective of this test is to determine the strength of cylindrical rock specimens subjected to triaxial compression. From the results of this test the values of internal friction and cohesion of the rock material can be calculated.

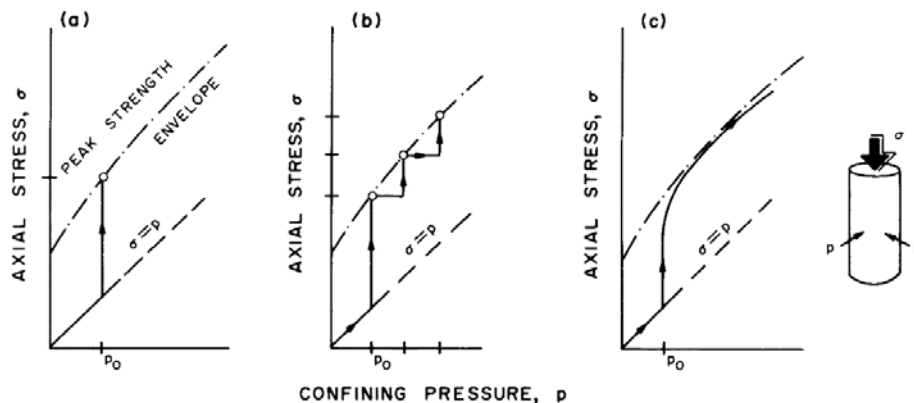


Fig.1: Different triaxial test types: (a) Type I - Individual test; (b) Type II - Multiple failure state test; (c) Type III – Continuous failure state test (ISRM 1983).

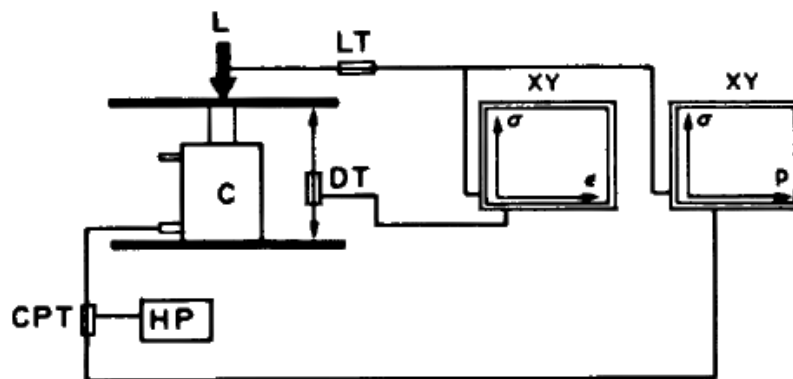
Three different types of triaxial compression test can be conducted in the laboratory. The tests are intended to measure strength of cylindrical rock specimens as a function of confining pressure. The three test types differ from each other in the manner by which the strength envelope is produced (Fig 1a – c). With test type I (“individual test”), individual points on the failure (peak strength) envelope are obtained from several tests (Fig.1a), while with test type II (“multiple failure state test”) (Fig.1b) and test type III (“continuous failure state test”) (Fig.1c) the envelope is produced with a single test employing a stepwise or continuous procedure. The type I (individual test) of triaxial compression test is described.

APPARATUS USED

- A triaxial cell body in which the specimen may be placed in order to apply the confining pressure.
- Both bearing plates shall be equipped with a spherical seat. The centres of curvature of the spherical seats must coincide with the centres of the end faces of the specimen.
- A flexible jacket to prevent the hydraulic fluid from entering the specimen. It shall be sufficiently rigid, so as not to enter surface pores.
- A testing machine of sufficient capacity and capable of applying controlling and measuring the axial load at a rate conforming to the requirements described in the test procedure. The steel platens with which the specimen is loaded shall consist of hardened steel (specifications see ASTM and ISRM) and be at least as large, but preferably not more than 2 mm larger than the diameter of the specimen. The spherical seating of the testing machine (if present) shall be removed or placed in a locked position.
- A hydraulic pump and oil pressure gauge, capable to maintain and measure the confining oil pressure with an accuracy of 2%.

PREPARATION OF THE TEST SPECIMENS

- Test specimens shall be right circular cylinders having a height to diameter ratio of 2.0 – 3.0 and a diameter preferable of not less than NX core size (54mm). The diameter of the specimen should be at least 10 times larger than the average grain size.
- The number of specimens to be tested, as well as the number of different confining pressure values should be determined from practical considerations. However, at least five specimens per rock sample are preferred in addition to the uniaxial compressive strength tests conducted according to the relevant specifications. If the sample rock is anisotropic due to the presence of weakness planes and/or preferred orientation of minerals, the specimens should be prepared in such a way that both directions parallel as well as perpendicular to such planes can be tested. If enough testing specimens can be prepared from the available sample, various intermediate angles could also be tested.
- The end faces of the specimen shall be flat to 0.02 mm and be perpendicular to the specimen axis within 0.25 (0.25 mm in 50 mm). During the test, capping of the samples is not permitted. The sides of the cylinder shall be smooth and free of abrupt irregularities and straight to within 0.5 mm over the full length of the specimen.
- The diameter of the specimen shall be recorded to the nearest 0.1 mm by taking two perpendicular measurements at three different heights of the cylinder. The height of the cylinder shall be determined to the nearest 0.1 mm.
- Moisture can have a significant effect on the triaxial strength of the test specimen. When possible the natural water content shall be preserved until the time of the test.



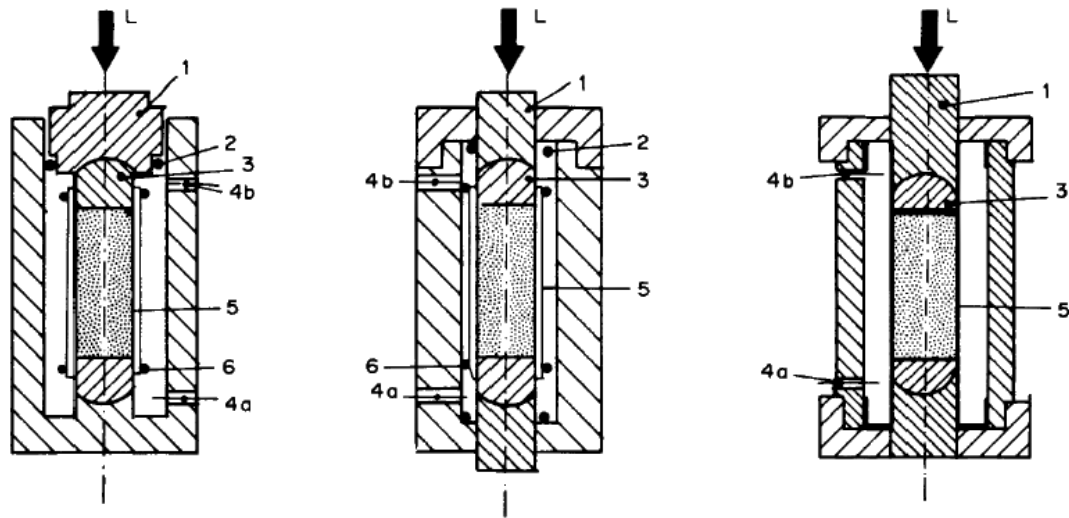
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|-----------------------------|--------------------------------------|
| L = Axial load. | CPT = Confining pressure transducer. |
| LT = Load transducer. | HP = Device for applying |
| C = Triaxial cell. | confining pressure. |
| DT = Deformation transducer | |
| (optional in test type I). | |

Fig.2: A schematic diagram of the test equipment.

TEST PROCEDURE

- The cell shall be assembled with the specimen aligned between steel platens and surrounded by the flexible membrane. The specimen, the platens and the spherical seats shall be accurately aligned so that each is coaxial with the others.
- The spherical seats shall be lightly lubricated with mineral oil.
- The cell shall be filled with hydraulic fluid, allowing the air to escape through an air bleeder hole. The air bleeder hole shall then be closed.
- The cell shall be placed into the axial loading device (Fig.2)

- e) The axial load and the confining pressure should be increased simultaneously and in such a way that axial stress and confining pressure be approximately equal, until the predetermined test level for the confining pressure is reached. Subsequently the confining pressure shall be maintained to within 2 percent of the prescribed value.
- f) The axial load on the test specimen shall then be increased continuously and without shock to produce an approximately constant rate of load or deformation; such that failure will occur within 5 to 15 minutes of loading. Alternatively the stress rate shall be within the limits of 0.5 to 1.0 MPa/s. The axial failure load shall be measured within 1 percent accuracy.
- g) The maximum axial load and the corresponding confining pressure on the specimen shall be recorded.
- h) The axial displacement shall be measured to an accuracy not less than 0.5 percent.



1 = Loading piston.
 2 = Seals.
 3 = Platens with spherical seating.
 4a = Hydraulic connection.
 4b = Bleeder hole.

L = Load applied by loading device.

5 = Flexible membrane.
 6 = "O"-ring clamps.

Fig.3: Three possible designs for triaxial chamber

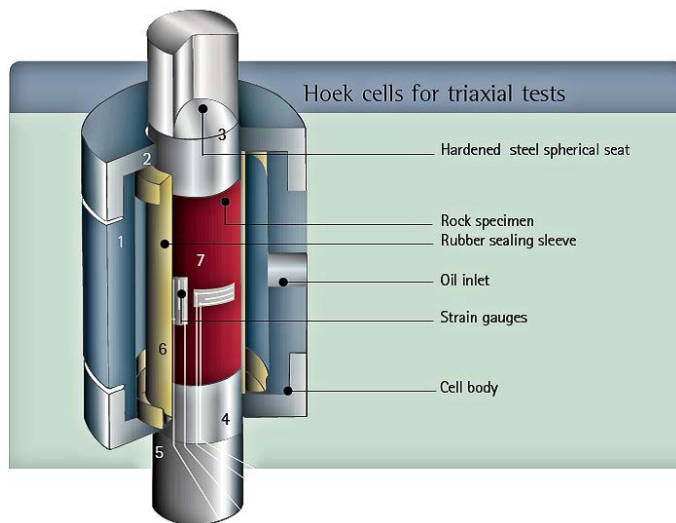


Fig.4: Hoek cell for triaxial tests

General description of Hoek cell

Hoek triaxial cells are offered in four models of different size; each one consists of the following:

- A cell body 1, complete with two quick release self-sealing couplings, for the introduction of the hydraulic oil and cell pressure, and for air outlet and saturation of the chamber respectively.
- Two end caps 2, screwed into the cylindrical cell body.
- Upper 3 and lower loading caps 4 with spherical coupling
- Two female spherical seatings 5 connected to the loading caps 3 and 4 for the correct transmission of the axial load from the loading frame of the compression machine to the specimen.
- A rubber sealing sleeve 6 to separate the specimen from the cell fluid.

Measurements of axial and radial strain are carried out with the use of electric strain gauges 7, both in vertical and horizontal direction, directly glued on the lateral surface of the specimen. The wiring connections are passed within the rubber sleeve through the cell body and loading cap. Each strain gauge must be connected to a proper electric device to complete and balance the Wheatstone bridge. The strain gauge measurements can be performed by automatic testing systems or by suitable Datalog.

OBSERVATIONS

Specimen no.	Average Specimen diameter (mm)	Average Specimen Length (mm)	Confining Pressure (MPa)	Axial Load (tonne/kg)	Axial Stress (MPa)

CALCULATIONS

- a) The compressive strength of the specimen shall be calculated by dividing the maximum axial load, applied to the specimen during the test, by the original cross sectional area of the specimen.
- b) The confining pressures and the corresponding strength values for different specimen are plotted with the confining pressures as abscissa and strength as ordinates (Fig.5).
- c) A strength envelope is obtained by fitting a mean curve to the above points. From practical considerations it may be advisable to fit a straight line to only the most relevant part of the curve, or to fit several straight lines to different parts of the curve. Each straight line is characterized by calculating its gradient m (tangent of the inclination) and its Y intercept, b on the vertical (axial stress) axis (Fig. 5).
- d) Using parameters m and b , the angle of internal friction ϕ and a value for the apparent cohesion c (in the sense of coulomb's failure theory) may be calculated using the formulae:

$$\phi = \arcsin \frac{m - 1}{m + 1}$$

$$c = b \sin \frac{1 - \sin \phi}{2 \cos \phi}$$

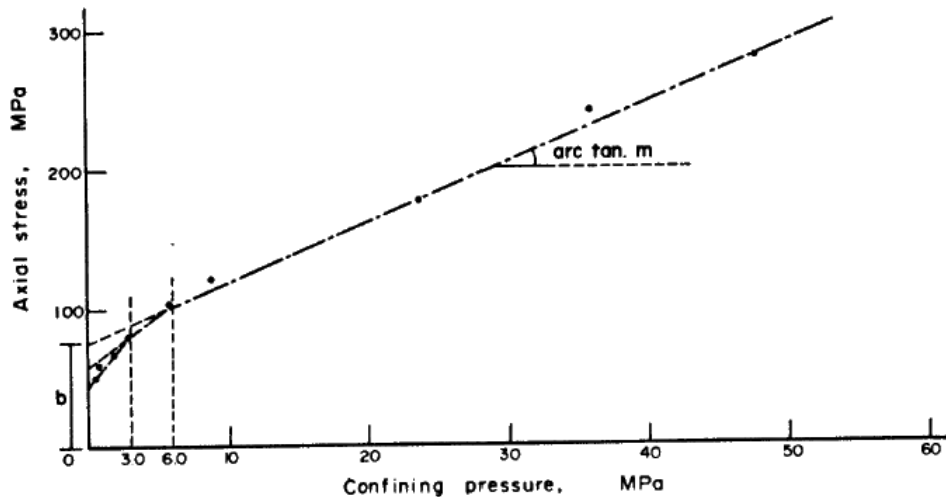


Fig. 5: Strength envelope

REPORTING OF RESULTS

The report shall include the following information:

(a) Data on the sampling:

- Project name, location, date of sampling, sample number, depth below terrain (in case of borehole)
- Type of sample (core, block, disturbed, or other), sample dimensions
- Lithology, weathering grade, grain size, natural water content
- Sample transport and storage conditions

(b) Data on the specimens:

- Form, dimensions and weight of all specimens
- Density and water content during testing

(c) Data on the testing procedure:

- Testing equipment used (loading frame, triaxial cell equipment to create and measure confining pressure)
- Loading rate during the test, sketch of failure.
- Table with specimen number, specimen height, specimen diameter, confining pressure and the corresponding axial strength.
- The plot of axial strength vs confining pressure (Fig.5)
- A table giving the values of C and ϕ together with the range of confining pressures in which they are valid.

REFERENCES

International Society for Rock Mechanics Suggested Methods "Rock Characterization, Testing and Monitoring" Editor E.T. Brown, Pergamon Press 1981, pages 125 to 127.

Suggested methods for determining the strength of rock materials in triaxial compression: Revised version, *Int. J. Rock Mech. Min. Sci. & Geomech. Abstr.* Vol. 20, No. 6, pp. 283-290, 1983.

Experiment-9

Name of the experiment: Determination of Cerchar Abrasivity Index of rocks

Scope: The CERCHAR Abrasivity Test is intended as an index test for classifying the abrasivity of a rock material.

Apparatus:

1. Stylus: It is strongly recommended to employ only styli tempered to Rockwell hardness HRC 55 ± 1 . The diameter of the stylus should be at least 6 mm. The tip of a stylus shall have a conical angle of 90° . A worn stylus should be re-sharpened and the tip angle checked under a microscope before use in a further test.
2. Force: The static force acting on the stylus should be 70 N.
3. Grinder: Each used stylus should be re-sharpened using a standard abrasive stone wheel.
4. Test Specimen: The rock sample may be either disc-shaped or irregular in shape. The size of the rock surface should be sufficient to permit five test scratches that are at least 5 mm from the edge of the rock surface. Each test should be 5 mm apart.



Test Procedure:

1. The sample should be clamped firmly in the vice while observing the desired scratching direction.
2. The rock surface should be, to the extent possible, horizontal.
3. The stylus should be carefully lowered onto the rock surface to avoid any damage to the tip of the stylus.
4. The stylus should be positioned so it is vertical and perpendicular to the rock surface.
5. The length of a test scratch in the rock sample must be exactly 10.0 mm within 1 s.
6. During the test there should be constant contact between the stylus and the rock surface.
7. After testing, the stylus is carefully lifted from the rock surface and the stylus removed.
8. A minimum of five test replications must be made on the rock surface, each time by a new or re-sharpened stylus.

Stylus Wear Measurement:

1. The length or diameter of the wear flat, d , shall be based on optical and digital methods using a microscope having a minimum magnification of 25X.
2. The measuring resolution should be at least ± 0.005 mm with readings reported to the nearest 0.01 mm.
3. When using the side-view method, it is suggested the stylus should be placed in a V-notch holder or jig and four measurements shall be made each at 90° rotation.
4. The measurements should be taken parallel and perpendicular to the direction of scratching.

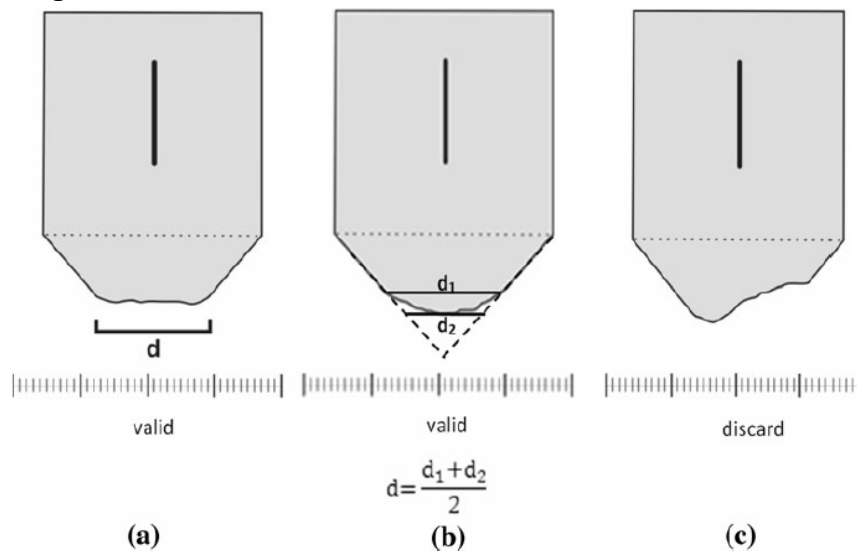


Fig. Standard worn profiles (a, b) and the corresponding length of wear surface, (c) an example of a non-standard profile in which case no measurement should be recorded.

Calculations:

For each measurement of the wear flat, d , the CAI is calculated by the formula given in Eq. (1)

$$CAI = d \times 10 \quad (1)$$

where d is the wear tip surface measured to an accuracy of 0.01 mm.

The dimensionless CAI value is reported as the arithmetic mean of five or more test replications.

Classification of CAI:

Mean CAI	Classification
0.1–0.4	Extremely low
0.5–0.9	Very low
1.0–1.9	Low
2.0–2.9	Medium
3.0–3.9	High
4.0–4.9	Very high
≥ 5	Extremely high

Data Table:

SN.	Rock Sample	d ₁	d ₂	d ₃	d ₄	Avg. dia (d)
1						
2						
3						
4						
5						

Results:**Precaution:**

1. Height of sample should be enough to keep the loading axis perfectly vertical so that eccentricity does not occur.
2. Rock surface to be tested should not be grinded after saw cutting.

Experiment-10

Name of the experiment: Determination of strength of concrete/rock using Schmidt hammer

INTRODUCTION

The Schmidt hammer test was developed in 1948 for non-destructive testing of concrete hardness, and was later used to estimate rock strength. With its portable, simple and affordable attributes, the Schmidt hammer (SH) is an ideal index apparatus, which underlies its increasing popularity and expanding range of applications. The SH rebound hardness value (R) is perhaps the most frequently used index in rock mechanics practice for estimating the uniaxial compressive strength (UCS) and the modulus of elasticity (E) of intact rock both in laboratory conditions and in situ. The SH is also widely used for estimating the UCS of discontinuity walls and assessing the workability, excavability and boreability of rocks by mechanical means (cutting, polishing, milling, crushing and fragmentation processes in quarrying, drilling and tunneling). Researchers have sought to establish correlations between the SH rebound values (R) and the UCS and E for different rock types.

SCOPE

The use of the SH to determine the rebound hardness of rock surfaces both in laboratory conditions and in situ with an emphasis on the use of this hardness value as an index of the UCS and E of rock materials.

APPARATUS

a) Operational principle

Schmidt hammer apparatus is used for obtaining rebound hardness value of rock. The mechanism of operation is simple: a plunger released by a spring impacts against the rock surface, and the rebound distance of the plunger is then read directly from the numerical scale or electronic display ranging from 10 to 100 (Fig. 1). Schmidt hammer models are designed with different levels of impact energy. The standard L- and N-type Schmidt hammers (LR and NR Type with SH recording facilities) are built to generate different levels of impact energy: 0.735 and 2.207 Nm, respectively. Both types are commonly used to determine rock properties. These are used with caution when the UCS of the rock material or discontinuity wall is outside the range of 20–150 MPa, where sensitivity decreases and data scatter increases. The N-type hammer is less sensitive to surface irregularities, and is preferred in field applications; while the L-type hammer has greater sensitivity in the lower range and gives better results when testing weak, porous and weathered rocks. The higher impact energy of N-type hammer (corresponding to probing a larger volume of material by a deeper and wider penetration) reduces scatter in rebound values compared to L-type hammers.

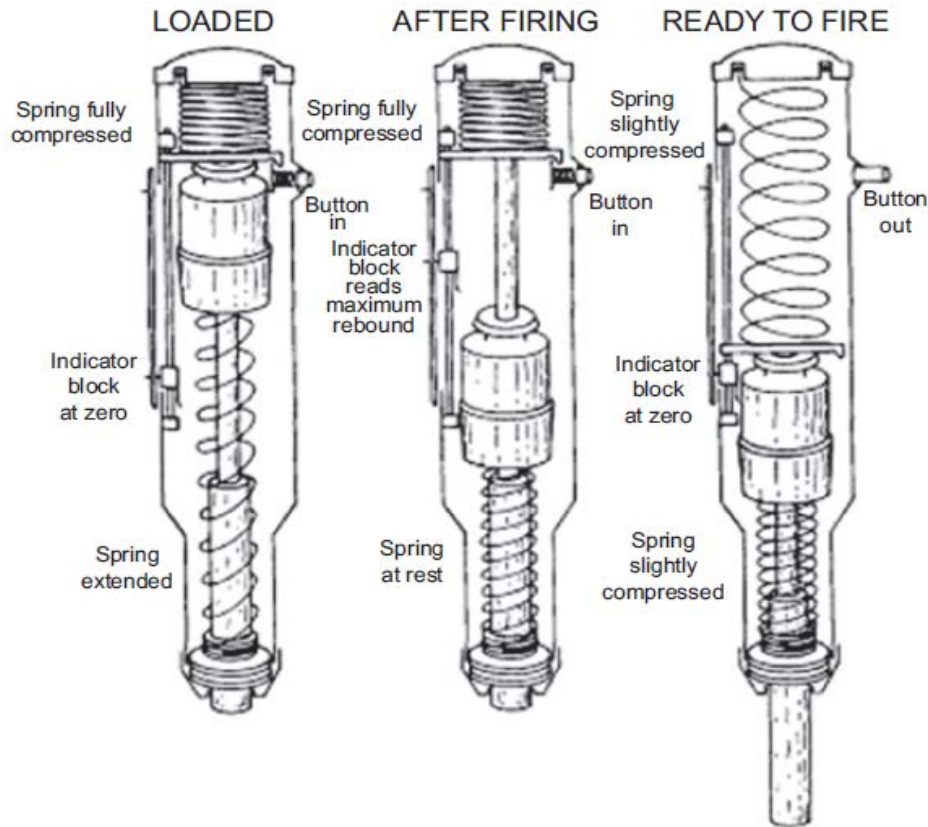


Fig.1. Working principle of a Schmidt hammer

b) Calibration

SH are supplied with calibration anvils with vertically guided impact points made of steel as hard as that of the plunger tip (usually Brinell 500 or Rockwell 52 C). It is essential to verify that the hammers maintain their standard rebound values before and after field investigations. In correlation studies, two consistent readings within the predetermined range of rebound from the anvil should be taken before and after testing each specimen. A drift in the calibrated rebound values may suggest that the key spring is losing its stiffness and should ideally be replaced. If this is not possible, a correction factor (CF) for the hammer should be calculated and applied to all readings to account for the loss of stiffness:

$$CF = \frac{\text{specified standard value of the anvil}}{\text{average of ten readings on the anvil}}$$

SPECIMEN REQUIREMENTS

The Schmidt hammer test can be carried out on core samples; unfractured rock blocks and fractured rock mass (Fig. 2). In the laboratory, core samples or unfractured rock blocks can be tested. In the field, Schmidt hammer test can be performed on large rock blocks and on fractured rock mass. The test results conducted on fractured rock mass will be different from the other test results because of the fracturing.

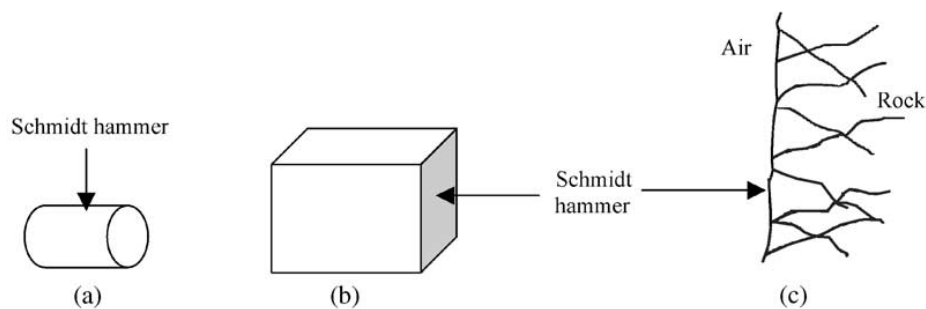


Fig.2. The application types of Schmidt hammer test: (a) core, (b) unfractured rock block, and (c) in situ fractured rock mass

Specimens should be intact (free of visible cracks), petrographically uniform and representative of the rock mass domain (identified from cores or exposures) being characterized. Test surfaces, especially under the plunger tip (impact points), should be smooth and free of dust and particles. In the field, a medium-grained abrasive stone can be used for local smoothing of rough surfaces in hard rock. Fine sandpaper can be used to smooth the surfaces of cores and block specimens, especially when drilling or sawing produces visible ridges. Cores and blocks should be air dried or saturated before testing. When this is not possible, the degree of moistness of the surface and the specimen as a whole should be recorded as wet, moist or damp. Cores should be of at least NX size (≥ 54.7 mm) for the L-type hammer and preferably T2 size (≥ 84 mm) for the N-type. Block specimens should be at least 100 mm thick at the point of impact. It is essential that impact energy is not dissipated in the form of wave scatter or cracking because the impact points are too close to the specimen boundaries.

TEST PROCEDURE

When conducting the test the hammer should be held at right angles to the surface which in turn should be flat and smooth. The pressure on the plunger of the hammer should be gradually increased until the hammer impacts. After the impact, the rebound number should be recorded. The rebound reading gets affected by the orientation of the hammer. When it is used in vertical position gravity will increase the rebound distance of the mass and vice versa for a test conducted on a floor. So, it is necessary to record the orientation of hammer while conducting the test.

Twenty rebound values, as recommended by the earlier ISRM suggested method should be recorded from single impacts separated by at least a plunger diameter. In order to provide similar degrees of confinement in all directions, impact points should be one radius away from the nearest end of core specimens and half the thickness away from block boundaries. Examine the impression should made on the surface after the impacts and disregard the reading if impact crushes or breaks the surface.

REPORTING OF RESULT

ISRM suggested that the average of the upper ten values out of 20 rebound values obtained should be taken for calculating Schmidt hammers number. Schmidt hammers number readings have been used to estimate compressive strength of rocks. Rebound numbers have been correlated with core/cube testing values. The correlation graph between rebound number and strength is normally provided by the rebound manufacturer as shown in Figure 3 having four curves each one representing the orientation hammer at the time of testing. The correlation graph should be used for estimating compressive strength of the rock using the respective curve. However, empirical relationship established by various researchers is also used for

estimating compressive strength as well as the other mechanical properties of rock from the obtained Schmidt hammers number. Average Schmidt hammers number of a few types of rock is given in Table 1.

Table 1: Average Schmidt hammer values of some rocks

Schmidt hammer SH - values	Rock type
50-60	diabase, serpentinite, quartzite
40-50	basalt, amphibolite, gabbro, gneiss, granite
30-40	marble, limestone, sandstone, slate
15-30	schist, siltstone
<15	chalk

The test report should include the other following information:

- Lithological description of the rock (e.g. color, texture/fabric, weathering/ alteration, rock name with grain size)
- Geographical location and depth of sampling or in-situ rock faces
- Date of sampling or excavation and testing, and storage conditions or climate
- Specimen or face number
- Method of excavation or block production (e.g. blasting, ripping, mechanical splitting, boring)
- Dimensions of specimens or exposure surfaces
- Sample moisture during testing (water content % or in descriptive terms such as dry, moist, damp)
- Hammer type (L-, N- or another type)
- Orientation of hammer axis (impact direction) with reference to horizontal(in degrees, down ward being $+90^0$ and upward -90^0)
- Orientation of hammer axis with reference to intact rock anisotropy features (e.g. lamination, foliation, schistosity, lineation)



Fig.3. Schmidt hammer

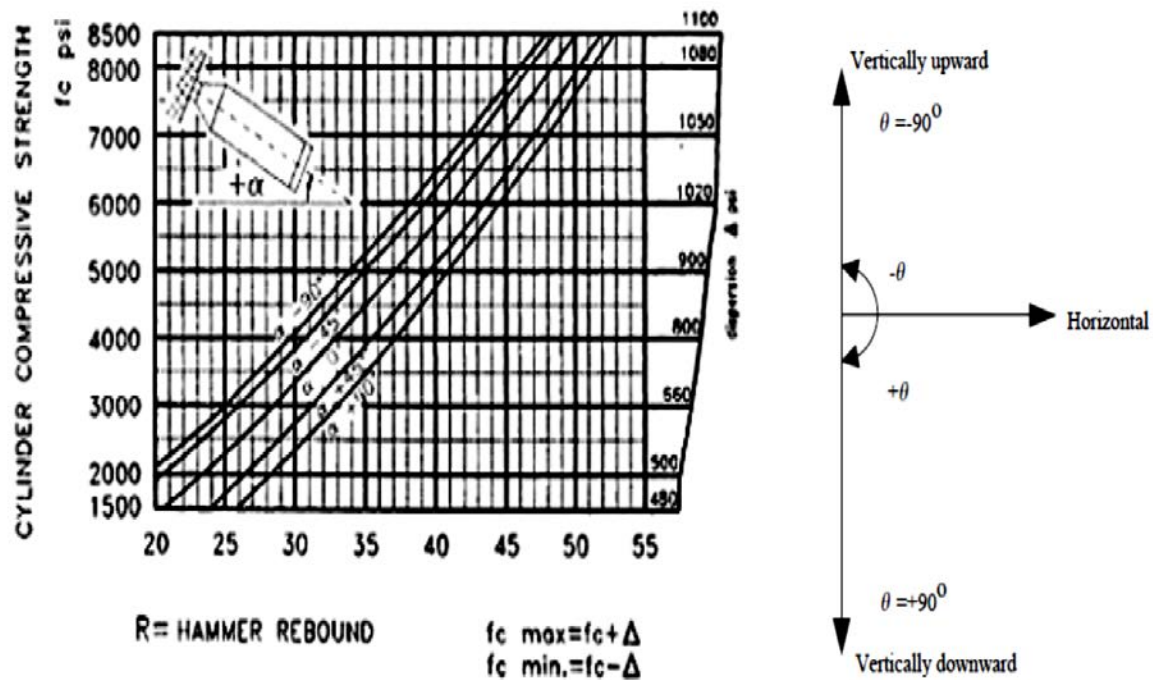


Fig.3: The correlation chart between rebound number and strength

Influencing parameters

The test results are sensitive to the following factors:

- *Relative strength of coarse grains vs. matrix:* The size and distribution of grains and the relative strength of the matrix has a considerable influence on the degree of scatter of rebound values.
- *Weathering and moisture content:* Microstructural changes induced by weathering result in different response mechanisms, especially in crystalline igneous rocks, and significantly different rebound values.
- *Moisture content:* Moisture content of the rock within the zone of influence of impact may considerably affect the rebound values depending on its microstructural character.
- *Anisotropy:* Planes of anisotropy in laminated and schistose rocks such as shale, slate, phyllite and schist control the response to impact and loading.
- *Field vs. laboratory testing:* In most cases, the degree of scatter will increase and the average magnitude of rebound values will decrease in field testing compare to that of laboratory test.
- *Hammer type:* The intensity of impact energy of type of hammer may produce different rebound value.
- *Orientation of hammer axis:* The impact direction of hammer with respect to horizontal as well as the orientation of anisotropy of rock significantly influence effects the SH value.

Careful observation of the parameter mentioned above should be made while carrying out the study for obtaining reliable test result.

Experiment-11

Name of the experiment: Determination of In-situ stress measurement using hydro-fracturing technique

INTRODUCTION

Hydraulic fracturing (HF) is boreholes field-test method designed to assess the state of in situ stress in the earth crust through a drill hole. This method is also referred to as hydrofracturing, or hydrofrac, and sometimes as minifrac. The HF stress determination method derives from a technique originally developed by the petroleum industry to stimulate oil production by increasing the overall porosity and permeability of rock. HF test results generally in an estimate of the state of in situ stress (both magnitudes and directions) in the plane perpendicular to the axis of the borehole. When both the borehole and the induced HF are nearly vertical, the stress component in the direction of the hole axis is taken as being principal and equal to the overburden weight ($\rho g h$). The domain of application of the HF method has been extended with the HTPF (Hydraulic Testing on Pre-existing Fracture) method. The HTPF method provides an evaluation of the complete stress tensor (6 components), independent of borehole orientation and material properties. When possible, both methods should be combined for optimum results. HF is the only rock stress determination technique that has been successfully applied to deep drill holes. Thus, this technique has found application in site characterization investigations via vertical drill holes from the surface.

APPARATUS

1. Minifrac system

Module I: Installation & Test Tools comprising

8 x Installation rods (1.5 m each)

2 x Fracturing Tools

2 x Impression Tools

Test Pipe

Module II: Pressurisation and recording system comprising

Hydraulic tank with dual hand pumps

Instrument Panel with Test and Packer Manifolds

2 x Pressure gauges (40 MPa)

Recorder and Pressure sensors

2 x Vent Valves

2 x Test and Packer Hoses (5 m each)

Laboratory test frame

PRINCIPLE

For both HF and HTPF methods, a section of a borehole is sealed off by use of two inflatable rubber packers sufficiently pressurized so that they adhere to the borehole wall. Hydraulic fluid (typically water) is pumped under constant flow rate into the section, gradually raising the pressure on the borehole wall until a fracture is initiated in the rock, or a pre-existing fracture

is mechanically opened. Pumping is stopped, allowing the interval pressure to decay. Several minutes into the shut-off phase, the pressure is released and allowed to return to ambient conditions. The pressure cycle is repeated several times maintaining the same flow rate. Key pressure values used in the computation of the in situ stresses are picked from the pressure–time record. The repeated cycles provide redundant readings of the key pressures. The attitude of the induced HF, or of the pre-existing fracture, is obtained using an oriented impression packer or one of several geophysical logging methods. HF orientation is related to the directions of the principal stresses. With HF, data from the pressurization and fracture orientation phases of the test are used to obtain the in situ principal stresses in the plane perpendicular to the borehole axis. With HTPF, tests yield an evaluation of the normal stress supported by fracture planes with different known orientations, and the complete stress evaluation results from an inversion of these results.

The method is, in general, more suited to measurements at depths >50 m that are beyond the capabilities of most other techniques. It has the advantage of requiring no advance knowledge of the elastic properties of the rock and being able to be carried out without difficulty below the water table. It tends to measure stresses over a relatively large area, >0.5 - 1.0m dia, not at a point. The method is most accurate when applied in materials whose behaviour approaches that of brittle, homogeneous, elastic, isotropic and non-porous media.

APPARATUS

The same apparatus is necessary for HF and for HTPF, see Figs. 1(a, b).

Surface equipment: A sturdy tripod or a drilling rig is placed over the borehole collar for tripping the downhole tools necessary for conducting the tests. When drill pipe or steel tubing is used for lowering the tools, a drilling rig is preferred because it can accommodate the heavy weight of the downhole assembly. A tripod may be used when tools are lowered on the much lighter wireline.

Straddle packer: Sealing of the borehole test interval is accomplished by use of two inflatable rubber packers, spaced apart a distance equal to at least six times the hole diameter. The two packers are connected mechanically as well as hydraulically to form one unit termed the straddle packer.

High-pressure tubing, drill pipe, or hose: Generally, rubber packer inflation and test interval pressurization are carried out hydraulically from the surface, although for deep tests often associated with the petroleum industry, equipment exists that employs a pump directly attached to the straddle packer and is remotely controlled. Hydraulic fluid is conveyed downhole typically through the use of high-pressure stainless steel tubing, flexible hose, or drill pipes (also called drill rods). When drill rod is used, it also serves for tripping the hydrofracturing equipment. Tubing or hose require an additional means of lowering and lifting the test tools, such as drill rod or wireline.

Pressure gages, pressure transducers, and flow meter: Pressure gages are used on the surface to give visual real-time information of the hydraulic fluid pressure. Pressure transducers are used to monitor and transmit pressure data to a recording device. In some setups, only the test interval pressure is monitored. The preferred arrangement is one in which the packers and the interval are pressurized and monitored independently. For shallow tests, surface transducers are sufficiently accurate. For depths exceeding several hundred meters, downhole pressure

transducers (or transmitters) emplaced close to the location of the packer are preferred. These provide a more accurate recording of the test interval pressure. A flow meter is employed to monitor the flow of fluid into the test interval. Typically this is a surface device.

Pressure generators: Most often, hydraulic fluid pressure is provided by a pump or pumps located on the surface. However, for deep tests, equipment exists for which the pump is fixed on the straddle packer. Some pumps are capable of providing up to 100MPa at a typical flow rate range of 1–10 l/min. The pump is powered electrically, pneumatically, or by use of a combustion or diesel engine.

Recording equipment: Analog data from the pressure transducers and flow meter are fed into a computer data acquisition program via an analog/digital board. Separate analog real-time reading of the test interval and packer pressures and of the flow rate are often provided by a multi-channel strip-chart recorder.

Fracture Orientation Detection Devices

Impression packer: An image of the borehole wall within the test interval is commonly obtained using an impression packer, an inflatable packer with an outer layer of very soft semi-cured rubber. This packer is inflated when it is positioned precisely at the same depth as the HF test interval, resulting in an impression of the borehole wall and any fractures that traverse it.

Orienting tool: Attached to the impression packer is an orienting tool, which can be magnetic or gyroscopic. The magnetic tool enables a camera to photograph the position of magnetic north on the borehole wall, from which the orientation of any induced fractures can be obtained. It is considerably easier and less expensive to use than a gyroscopic tool, but is of little use in magnetic rocks (such as basalt, some gneiss, and others). Geophysical tools, such as borehole cameras, borehole viewers (a sonic device) or electrical imaging systems are also available for fracture orientation determination. All these fracture orientation techniques have their own advantages and limitations. However, when the electrical imaging system is directly mounted on the straddle packer, the same tool provides initial borehole reconnaissance (no need for cores), exact positioning at selected depth intervals, real-time imaging of fracture opening, and combined electrical and hydraulic signature of fracture opening and closing.

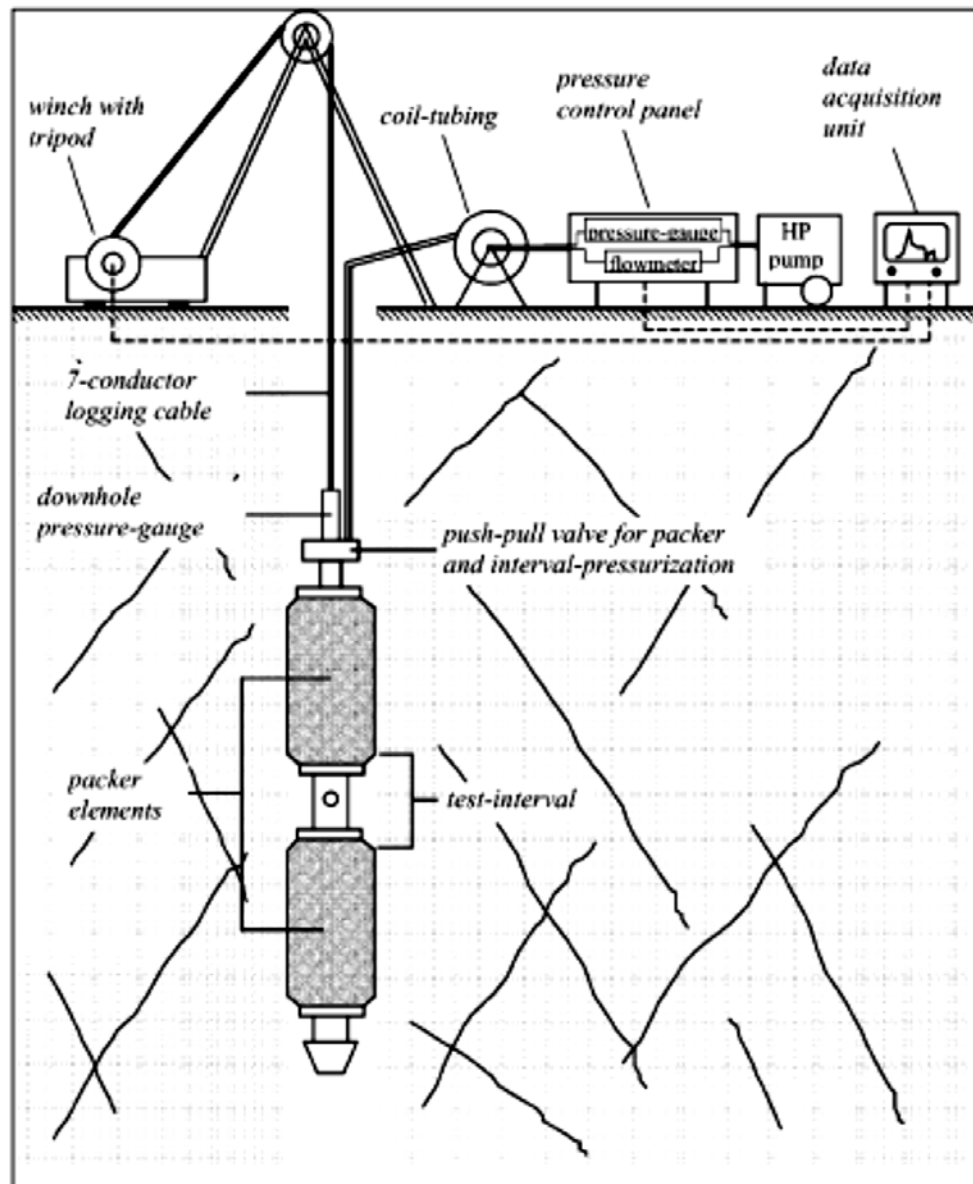


Fig. 1: (a) Schematic view of a hydraulic fracturing system (from Rummel et al.).

HYDRAULIC FRACTURING TEST PROCEDURE

During any test, continuously record the test interval and straddle packer pressures, the instantaneous injection flow rate and the total injected volume. When possible, keep track of the various vented volume. The following procedure is illustrated by Figs. 1 and 2.

1. Drill borehole (generally the vertical direction is recommended in the absence of topographical or structural effects), and extract continuous core or obtain clear and oriented borehole images. Depending on project and rock type, typical borehole diameters range from 76 to 96mm (N- to H-size) in site investigations for underground civil structures, and may extend to 180mm in deep petroleum, geothermal or scientific wellbores. Core or borehole images are essential for selecting intact test intervals and for identifying the rock formations to be tested. Cores are also necessary for any additional laboratory tests that may be useful for complementing the field tests. However, for deep boreholes, preference is given to rotary drilling associated with

geophysical imaging as a less expensive alternative to diamond drilling and coring.

2. For HF, select test zones that are devoid of fractures or other disturbances and are at the appropriate depths as per project requirements. For HTPF, select well-isolated planar fractures with dip and azimuth sufficiently variable to sample properly the state of stress. Extracted core and geophysical logs (such as caliper, sonic, density, televiwers or electrical imaging) are all useful for this purpose.
3. Seal off the test interval by positioning the straddle packer at the planned depth, and pressurize the packers to a typical level of 2–4 MPa. Such pressure will anchor the packers to the borehole wall without creating any fracturing (unless the rock is unusually weak). The hydraulic fluid is generally water, although alcohol may be preferred in some instances.
4. Pressurize the test interval for an initial qualitative permeability inspection (slug test). A downhole valve is preferred for a more accurate relation between the pressure decay and the local rock permeability.
5. Raise the interval pressure maintaining a constant, predetermined, flow rate. Note that the flow rate may depend on rock permeability and that the aim is to raise the interval pressure steadily so that the peak pressure at which rock at the borehole wall fractures (termed breakdown pressure), or the pre-existing fracture opens, is reached in 1–3 min. The hydraulic fluid may be adjusted to the local material characteristics, in particular in clayish or salty environments. With surface pumps, the fracturing fluid is conveyed to the test interval through a separate string of tubing or hose. In some cases, one single string for both packer inflation and test interval is used, and the destination is controlled by a downhole valve. Throughout pressurization, the packer pressure is maintained at about 2MPa higher than the interval pressure to ensure isolation. This can be best monitored and controlled if separate hydraulic lines and pumps are used.
6. Upon reaching breakdown pressure (or fracture opening), stop pumping but do not vent. Interval pressure will decay, first at a fast pace while the HF is still open and growing, and then at a much slower pace, after the fracture has closed. The pressure at which the fracture closes is termed shut-in pressure. A few minutes (typically 3–10 min) after shut-in the hydraulic line is vented.
7. An optional step here is that after venting a small volume (typically half a liter), the test interval is closed back and the rise in pressure is monitored. When the initial permeability test has shown that the rock is fairly impervious, this operation provides a means to verify that the fluid has been injected into the rock mass and that no significant bypassing to the borehole has occurred. However, in very porous and permeable environments, no rise in pressure may be observed. In such cases, it may be advisable to verify that injected flow rates are large enough to open the tested fracture for long enough distances. Finally, depending on test objectives, pressure is allowed either to reach ambient level if the step in Paragraph 8 is to be undertaken (which may take from a few seconds up to 30 min depending on site conditions), or the step in Paragraph 9 and/or Paragraph 10 is conducted immediately.
8. Fracture reopening-option 1 (for HF method only): After the pore pressure has reached its original value, i.e. after waiting for a few minutes, repeat the above pressurization cycle at least three times, using the same flow rate. The flow rate should be sufficiently

high to prevent fracturing fluid percolation into the closed fracture before the actual mechanical fracture opening. The additional cycles yield additional shut-in pressure values as the fracture is extended by the additional pumping. The peak pressures are discernibly lower than in the first pressure cycle, because the reopening of the fracture does not require overcoming the tensile strength of the rock. This yields the fracture reopening pressure. The fracture reopening pressurization cycles are recommended only for the HF method. They should not be undertaken with the HTPF method in order to limit chances of creating a real HF when a pre-existing fracture is being tested.

9. Fracture reopening-option 2 (for HTPF and HF methods): After the step in Paragraph 8 for HF, or directly after the venting and flow back of the step in Paragraph 7 for HTPF, conduct a step-rate pressurization cycle, in which the flow rate is first brought to a very low level and maintained constant while the pressure increases until it reaches a plateau. Thereafter, the flow rate is raised to a new step and again the pressure is allowed to equilibrate at a constant level, and this is repeated several times, yielding an array of constant pressure levels obtained at different flow rates. Typically, each constant pressure step lasts about 5 min. Once the fracture is fully open, injection stops and the test interval is sealed off for pressure drop monitoring. This provides an additional shut-in pressure reading. With the HTPF method, this cycle is conducted at least twice but possibly more, so that, combined with the step in Paragraph 6, a minimum of three shut-in pressure readings is produced.
10. Fracture reopening-option 3 (for HTPF and HF methods): This option is similar to option 2, except that, after reaching the maximum injection flow rate, the pressure is decreased progressively in a stepwise manner, so as to reach complete closure of the fracture. This closing process is taken to advantage for an additional independent shut-in pressure determination. It complements efficiently the step in Paragraph 9.
11. At the conclusion of the test, vent not only the test interval but also the packer pressure, to allow the packers to deflate so that they can be moved to the next testing depth.
12. Repeat the entire test, as described in Paragraphs 1–11, at all selected depth intervals.

Hydraulic-Fracture Delineation: Impression Packer Technique

The text here is illustrated by Figs. 2 (a, b)

1. When using the oriented impression packer to delineate the induced fracture on the borehole wall, trip the packer to the depth where its center coincides with that of the HF test interval. Inflate the impression packer to a pressure just higher than the recorded shut-in level and maintain it there for about 30 min. This enables a slight opening of the fracture, allowing the penetration of the soft rubber enveloping the impression packer. The amount of time is calculated to ensure that the fracture imprint on the packer will be preserved permanently.
2. During the inflation period, a camera mounted in the orienting tool attached to the impression packer is triggered by a preset clock to take a photograph of the face of the tool's magnetic compass, and of a line aligned with a marker on the packer. Thus, the impression packer is oriented with respect to magnetic north, enabling the orientation of any logged fracture.
3. Upon the retrieval of the impression packer, the camera film is developed, and the image of the compass face, showing the direction of the packer marker with respect to magnetic north, becomes part of the test record. The imprints of the HF on the impression packer are traced

relative to the oriented marker on a wrapped around transparent plastic sheet. The traces can be digitized to facilitate statistical evaluation of the average strike and dip of the induced fractures.

STRESS CALCULATIONS

Hydraulic fracturing (HF)

1. The calculations of in situ principal stresses given hereafter are for vertical boreholes (commonly used for HF), and for tests yielding vertical fractures (both within $\pm 15^\circ$ or so). This corresponds to the case in which the vertical stress component acts along a principal direction. 2. Least horizontal principal stress magnitude and direction (σ_h). Vertical horizontal fractures are oriented perpendicular to the direction of the minimum horizontal principal stress. The shut-in pressure (P_s) is the pressure needed to equilibrate the fracture-normal stress, which in this case is σ_h .

$$\sigma_h = p_s \quad (1a)$$

The direction of σ_h is obtained directly from the azimuth of the HF:

$$\sigma_h \text{ Direction} = \text{Direction of normal to vertical hydraulic fracture} \quad (1b)$$

3. Largest horizontal principal stress direction and magnitude (σ_H).

This principal stress is calculated based on the assumption of linear elasticity and insignificant effect of fracturing fluid rock infiltration: In the absence of pore fluid in the rock mass, the maximum horizontal principal stress magnitude is given by Eq. (2a)

$$\sigma_H = T + 3\sigma_h - P_b \quad (2a)$$

Where T is the tensile strength of the tested rock, and P_b is the breakdown pressure.

4. In saturated rocks with low permeability, so that there is no percolation of the fracturing fluid in the

Formation before fracture opening, it is often assumed that pore pressure is unaffected by the state of stress and that Terzaghi's effective stress concept applies to tensile ruptures. In this case Eq. (2b) is employed:

$$\sigma_H - P_o = T + 3(\sigma_h - P_o) - (P_b - P_o) \quad (2b)$$

5. The maximum horizontal principal stress is perpendicular to the σ_h direction:

$$\sigma_H \text{ Direction} = \text{Direction of vertical hydraulic fracture strike} \quad (2c)$$

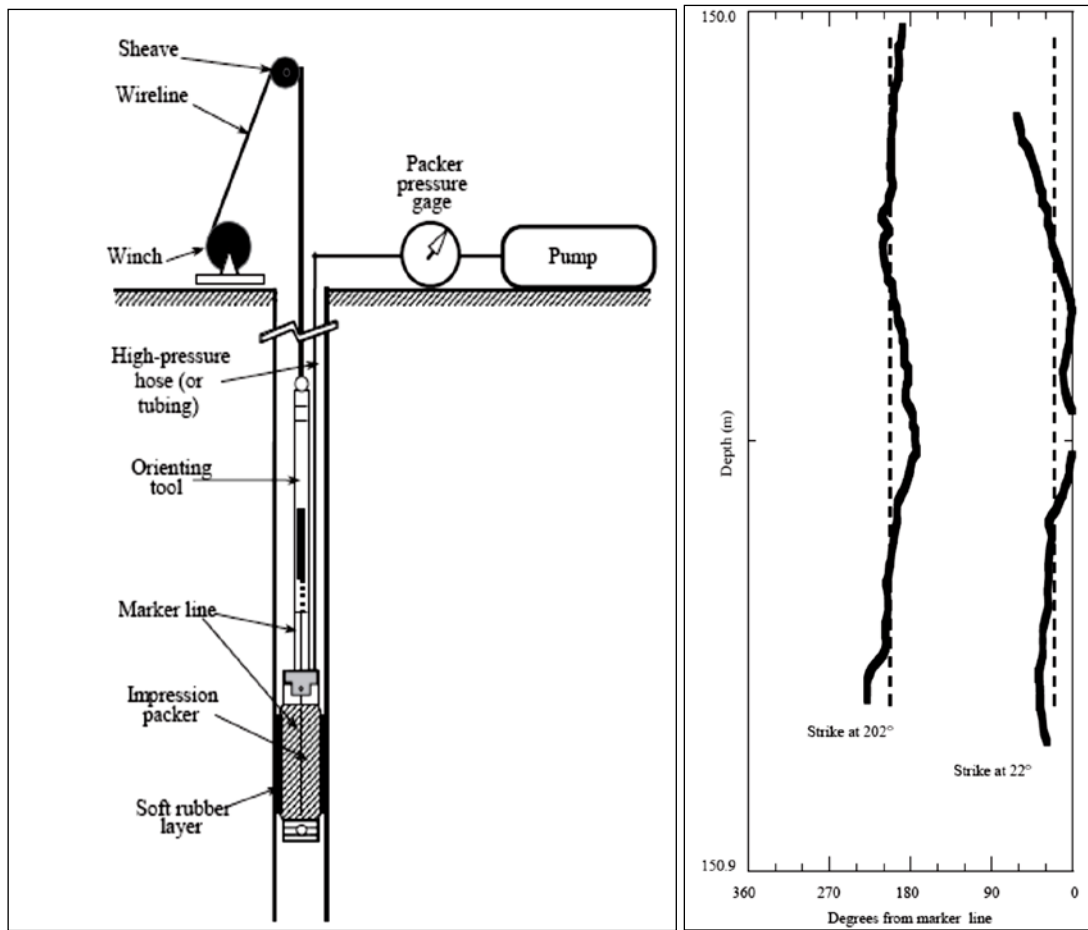


Fig.2: (a) Impression Packer Arrangement (b) section taken along the borehole wall by Impression packer

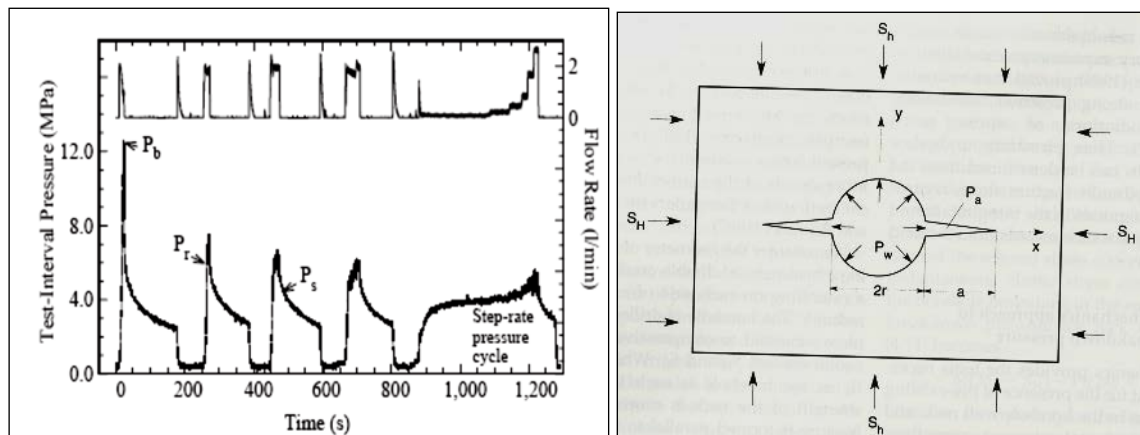


Fig. 3: (a) An actual record of test-interval pressure and flow rate versus time. The precise magnitudes of P_r and P_s are determined using techniques (b) Fracture Mechanics Model of HF in an infinite plate subjected to far field state of stress(after Rummel, 1987)

6. Whatever the role of pore pressure, solving Eqs. (2a) or (2b) requires that the rock tensile strength be known. The tensile strength can only be directly measured in the laboratory on core samples. The most common tensile test is the Brazilian test, which enables the testing of many disks cut directly from the extracted core. The Brazilian test configuration, however, does not

simulate conditions under HF, and the reliability of this test as representative of the tensile strength for HF has not been established. Core is also used to prepare hollow cylinders, which are fractured by applying internal pressure, with no external confining stress. This test accurately simulates an HF test in which there are no far-field stresses, and therefore the peak pressure is equal to the tensile strength T : The only unknown in such tests is the scale effect between field and laboratory dimensions.

7. When extracted core is not available, or laboratory tests are not feasible, or when tension tests appear to yield an unreasonable value for use in Eqs. (2a) or (2b), an alternative relation has been used, invoking the fracture reopening pressure (P_r). This pressure is assumed to be that at which the induced fracture, which has closed completely after initial pressure cycle, reopens. This time, however, fracture reopening does not have to overcome the tensile strength T ; and thus Eq. (2b) becomes

$$\sigma_H - P_o = 3(\sigma_h - P_o) - (P_r - P_o) \quad (3)$$

This equation for calculating σ_H has been widely used in field measurement campaigns. There is, however, considerable controversy regarding its reliability in cases such as when:

- (i) The induced fracture has not completely closed after each pressure cycle or the pore pressure has not returned to its original value,
- (ii) P_r has not been identified objectively on the pressure–time record,
- (iii) The volume of fluid pumped into the test interval is so large as compared to the intake of a slightly opened fracture that the correct P_r on the pressure–time curve may be missed

8. This is where the qualification and experience of the test personnel is particularly important in order to ascertain whether the picked value of the fracture reopening is a correct one. Hence, because of difficulties with both pore pressure effects and tensile strength estimation, the evaluation of the maximum horizontal principal stress magnitude involves a greater uncertainty than that of the minimum horizontal principal stress magnitude.

9. Vertical stress (σ_v): This component cannot be evaluated from test results unless the induced fracture is nearly horizontal, in which case the recorded shut-in pressure is taken as equal to the vertical stress. Otherwise, σ_v is assumed to be equal to the overburden weight per unit area at the depth of interest:

$$\sigma_v = \sum_{i=1}^n \rho_i g h_i \quad (4)$$

Where ρ_i is the mean mass density of rock layer i ; g is the local gravitational acceleration; h_i is the thickness of layer i ; and n is the number of rock layers overlying the test zone.



Fig 4: 1.Minifrac system



Fig 5: Module I: Installation & Test Tools



Fig 6: Module II: Pressurisation and recording system

Reference:

ISRM Suggested Methods for Rock Stress Estimation, International Society for Rock Mechanics Suggested Methods "Rock Characterization, Testing and Monitoring": 2007-14, Editor R. Ulusay

Experiment-12.A

Name of the experiment: Determination of Shear Strength of Soil Samples: Consolidated-Undrained Test

Introduction:

Shear strength in reference to soil is a term used to describe the maximum strength of soil at which point significant plastic deformation or yielding occurs due to an applied shear stress. The shear strength of soil with a maximum particle size of 4.75 mm is determined in unconsolidated undrained, consolidated undrained and consolidated drained conditions. Methods for determination of unconsolidated undrained and consolidated undrained tests are described below.

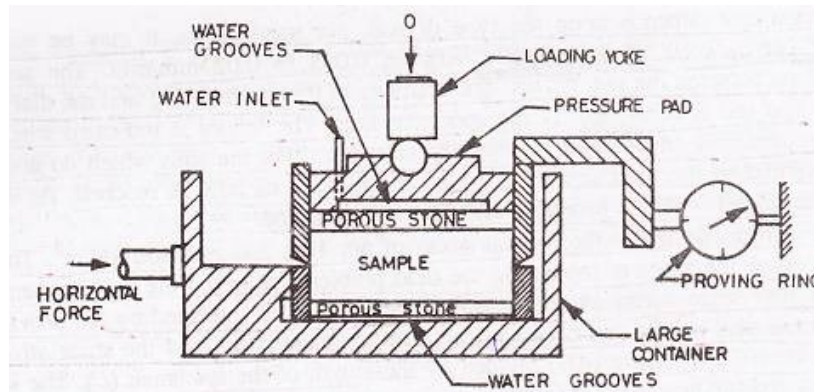


Fig. 1: Direct Shear Test Apparatus

EQUIPMENT

Soil shear apparatus and accessories consisting of

1. Shear box, divided into two halves by a horizontal plane, and fitted with locking and spacing screws
2. Box container to hold the shear box
3. Base plate having cross grooves on its top surface
4. Grid plates, perforated, 2 Nos.
5. Porous stones, 6mm thick, 2 Nos.
6. Loading pad
7. Loading frame: It shall satisfy the following requirements:
 - The vertical stress on the sample shall remain vertical and constant during the test and there shall be arrangement to measure compression.
 - The shear stress or strain can be applied in the dividing plane of the two parts of the shear box.
 - It shall be possible to maintain a constant rate of increase in stress during the test (irrespective of the strain rate) with arrangement to get different rates of stress increase.
 - In case of a strain-controlled apparatus, the strain rate should remain constant irrespective of the stress. Suitable arrangement shall be provided to obtain different strain rates.
 - No vibrations should be transmitted to the sample during the test and there should not be any loss of shear force due to friction between the loading frame and the shear box-container assembly.
8. Loading yoke
9. Weights - for providing the required normal loads

10. Proving ring-force measuring device fitted with a dial-gauge accurate to 0.002 mm to measure the shear force. Capacity: 100 to 250 kg capacity, depending on the type of soil.
11. Dial gauges, accuracy 0.01mm, 2 Nos. one suitably mounted to measure horizontal movement and the other suitably mounted to measure the vertical compression of the specimen.
12. Sample Trimmer or Core Cutter.
13. Balance - of 1 kg capacity, sensitive to 0.1 g.
14. Spatula and a Straight Edge
15. Stop Clock

PREPARATION OF SPECIMEN

- a) Cohesive soils may be compacted to the required density and moisture content, the sample extracted and then trimmed to the required size. Alternatively, the soil may be compacted to the required density and moisture content directly into the shear box after fixing the two-halves of the shear box together by means of the fixing screws.
- b) Cohesionless soils may be tamped in the shear box itself with the base plate and grid plate or porous stone as required in place at the bottom of the box.

The cut specimen shall be weighed and trimmings obtained during cutting shall be used to obtain the moisture content. Using this information, the bulk dry density of the specimen in the shear box shall be determined.

PROCEDURE

CONSOLIDATED UNDRAINED TEST

The apparatus should be assembled in a way similar to that given above except that instead of the plain grid plates, perforated grid plates and saturated porous stones should be used at the top and bottom of the specimen. The procedure is same as described unconsolidated undrained test except that after the application of normal stress, the vertical compression of the soil with time should be recorded.

The shear test should be conducted only after complete consolidation has occurred under the particular normal stress. The rate of shear should be such that water does not drain from the specimen at the time of application of the shear load. At the end of the test, the specimen should be removed from the box and the final moisture content measured. A minimum of three (preferably four) tests should be made on separated specimens of the same density at different normal stresses.

CALCULATIONS AND REPORT

Results of tests shall be recorded suitably. A recommended proforma for recording the results is given in the Appendix. From the calibration chart of the proving-ring, the loads corresponding to the load dial readings obtained during the test should be calculated. The loads so obtained divided by the corrected cross-sectional area of the specimen gives the shear stress in the sample. The corrected cross-sectional area shall be calculated from the following equation:

$$\text{Corrected area} = A_0 \left(1 - \frac{\delta}{3} \right)$$

Where

A_0 = Initial area of sample in cm^2 , and
 δ = displacement in cm.

The longitudinal displacement at a particular load may be either noted directly from the strain dial readings or calculated as the product of the corresponding time reading and the strain rate, allowing for the compression of the proving-ring, where applicable. The stress longitudinal displacement readings should be plotted and the maximum stress and corresponding longitudinal displacement together with the normal load applied during the test recorded.

The maximum shear stress and the corresponding longitudinal displacement and applied normal stress should be recorded for each test and the results should be presented in the form of a graph in which the applied normal stress is plotted as abscissa and the maximum shearing stress is plotted as ordinate to the same scale. The angle which the resulting straight line makes with the horizontal axis and the intercept which the straight line makes with the vertical axis shall be reported as the angle of shearing resistance and cohesion intercept respectively. In the case of the consolidated undrained test, the load at which the specimen is consolidated and the consolidation characteristics as determined during the consolidation part of the test should also be reported.

APPENDIX

Soil Specimen Measurements

Dimensions Area of specimen

Thickness of specimen

Volume of specimen

Initial wet weight of specimen

Moisture content (Average of tests)

Bulk density

Final wet weight of the specimen

Moisture content at shear zone

Consolidation (in Consolidated Undrained Test)

Applied load

Normal stress

Shearing Stage

Rate of shearing:

Sample No.	Displacement Dial reading	Displacement	Area Correction	Corrected Area	Stress Dial Reading	Shear Force	Shear Stress	Vertical Dial Reading	Vertical Dial Difference	Thickness of Sample

Plot shear stress-shear displacement curve and find:

- Maximum shear stress, and
- Corresponding shear displacement.

Sample No.	Normal Stress	Shear Stress at Failure	Shear Displacement at Failure	Initial Water Content	Final Water Content	Remarks

Plot shear normal stress displacement curve and find:

- Cohesion intercept, and
- Angle or shearing resistance.

Experiment-12.B

Name of the experiment: Determination of Shear Strength of Soil Samples: Unconsolidated-Undrained Test

Introduction:

Shear strength in reference to soil is a term used to describe the maximum strength of soil at which point significant plastic deformation or yielding occurs due to an applied shear stress. The shear strength of soil with a maximum particle size of 4.75 mm is determined in unconsolidated undrained, consolidated undrained and consolidated drained conditions. Methods for determination of unconsolidated undrained and consolidated undrained tests are described below.

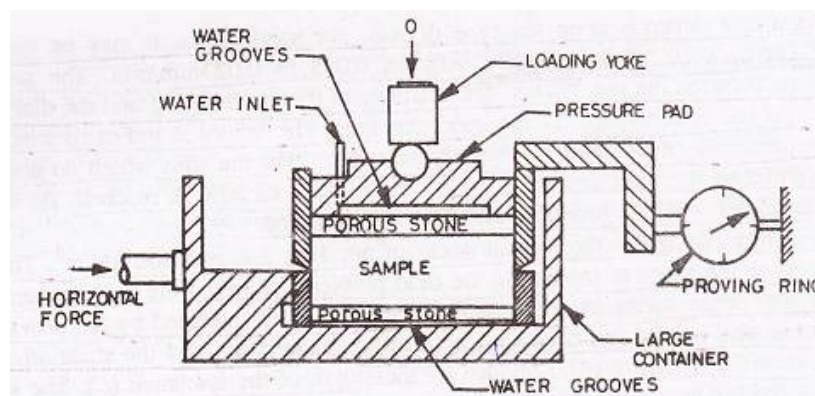


Fig. 1: Direct Shear Test Apparatus

EQUIPMENT

Soil shear apparatus and accessories consisting of

16. Shear box, divided into two halves by a horizontal plane, and fitted with locking and spacing screws
17. Box container to hold the shear box
18. Base plate having cross grooves on its top surface
19. Grid plates, perforated, 2 Nos.
20. Porous stones, 6mm thick, 2 Nos.
21. Loading pad
22. Loading frame: It shall satisfy the following requirements:
 - The vertical stress on the sample shall remain vertical and constant during the test and there shall be arrangement to measure compression.
 - The shear stress or strain can be applied in the dividing plane of the two parts of the shear box.
 - It shall be possible to maintain a constant rate of increase in stress during the test (irrespective of the strain rate) with arrangement to get different rates of stress increase.
 - In case of a strain-controlled apparatus, the strain rate should remain constant irrespective of the stress. Suitable arrangement shall be provided to obtain different strain rates.
 - No vibrations should be transmitted to the sample during the test and there should not be any loss of shear force due to friction between the loading frame and the shear box-container assembly.

23. Loading yoke
24. Weights - for providing the required normal loads
25. Proving ring-force measuring device fitted with a dial-gauge accurate to 0.002 mm to measure the shear force. Capacity: 100 to 250 kg capacity, depending on the type of soil.
26. Dial gauges, accuracy 0.01mm, 2 Nos. one suitably mounted to measure horizontal movement and the other suitably mounted to measure the vertical compression of the specimen.
27. Sample Trimmer or Core Cutter.
28. Balance - of 1 kg capacity, sensitive to 0.1 g.
29. Spatula and a Straight Edge
30. Stop Clock

PREPARATION OF SPECIMEN

- c) Cohesive soils may be compacted to the required density and moisture content, the sample extracted and then trimmed to the required size. Alternatively, the soil may be compacted to the required density and moisture content directly into the shear box after fixing the two-halves of the shear box together by means of the fixing screws.
- d) Cohesionless soils may be tamped in the shear box itself with the base plate and grid plate or porous stone as required in place at the bottom of the box.

The cut specimen shall be weighed and trimmings obtained during cutting shall be used to obtain the moisture content. Using this information, the bulk dry density of the specimen in the shear box shall be determined.

PROCEDURE:

UNCONSOLIDATED UNDRAINED TEST

The shear box with the specimen, plain grid plate over the base plate at the bottom of the specimen, and plain grid plate at the top of the specimen should be fitted into position in the load frame (the porous stones are not used for the undrained tests). The serrations of the grid plates should be at right angles to the direction of shear. The loading pad should be placed on the top grid plate. The water jacket should be provided so that the sample does not get dried during the test. The required normal stress should be applied and the rate of longitudinal displacement/shear stress application so adjusted that no drainage can occur in the sample during the test. The upper part of the shear box should be raised such that a gap of about 1 mm is left between the two parts of the box. The test may now be conducted by applying horizontal shear load to failure or to 20 percent longitudinal displacement, whichever occurs first. The shear load readings indicated by the proving ring assembly and the corresponding longitudinal displacements should be noted at regular intervals. If necessary, the vertical compression, if any, of the soil specimen may be measured to serve as a check to ensure that drainage has not taken place from the soil specimen. At the end of the test, the specimen should be removed from the box and the final moisture content measured. A minimum of three (preferably four) tests shall be made on separate specimens of the same density.

CALCULATIONS AND REPORT

Results of tests shall be recorded suitably. A recommended proforma for recording the results is given in the Appendix. From the calibration chart of the proving-ring, the loads corresponding to the load dial readings obtained during the test should be calculated. The loads so obtained divided by the corrected cross-sectional area of the specimen gives the shear stress

in the sample. The corrected cross-sectional area shall be calculated from the following equation:

$$\text{Corrected area} = A_o \left(1 - \frac{\delta}{3} \right)$$

Where

A_o = Initial area of sample in cm^2 , and

δ = displacement in cm.

The longitudinal displacement at a particular load may be either noted directly from the strain dial readings or calculated as the product of the corresponding time reading and the strain rate, allowing for the compression of the proving-ring, where applicable. The stress longitudinal displacement readings should be plotted and the maximum stress and corresponding longitudinal displacement together with the normal load applied during the test recorded.

The maximum shear stress and the corresponding longitudinal displacement and applied normal stress should be recorded for each test and the results should be presented in the form of a graph in which the applied normal stress is plotted as abscissa and the maximum shearing stress is plotted as ordinate to the same scale. The angle which the resulting straight line makes with the horizontal axis and the intercept which the straight line makes with the vertical axis shall be reported as the angle of shearing resistance and cohesion intercept respectively. In the case of the consolidated undrained test, the load at which the specimen is consolidated and the consolidation characteristics as determined during the consolidation part of the test should also be reported.

APPENDIX

Soil Specimen Measurements

Dimensions Area of specimen

Thickness of specimen

Volume of specimen

Initial wet weight of specimen

Moisture content (Average of tests)

Bulk density

Final wet weight of the specimen

Moisture content at shear zone

Consolidation (in Consolidated Undrained Test)

Applied load

Normal stress

Shearing Stage

Rate of shearing:

Sample No.	Displacement Dial reading	Displacement	Area Correction	Corrected Area	Stress Dial Reading	Shear Force	Shear Stress	Vertical Dial Reading	Vertical Dial Difference	Thickness of Sample

Plot shear stress-shear displacement curve and find:

a) Maximum shear stress, and

b) Corresponding shear displacement.

Sample No.	Normal Stress	Shear Stress at Failure	Shear Displacement at Failure	Initial Water Content	Final Water Content	Remarks

Plot shear normal stress displacement curve and find:

- a) Cohesion intercept, and
- b) Angle or shearing resistance.

Experiment-13.A

Name of the experiment: Determination of Consistency of Soil -Liquid & Plastic Limit

INTRODUCTION

The Atterberg limits are a basic measure of the nature of a fine-grained soil. Depending on the water content of the soil, it may appear in four states: solid, semi-solid, plastic and liquid. In each state the consistency and behavior of a soil is different and thus so are its engineering properties. Thus, the boundary between each state can be defined based on a change in the soil's behavior. The Atterberg limits can be used to distinguish between silt and clay, and it can distinguish between different types of silts and clays. These limits were created by Albert Atterberg, a Swedish chemist. They were later refined by Casagrande.

SCOPE

Liquid limit is significant to know the stress history and general properties of the soil met with construction. From the results of liquid limit the compression index may be estimated. The compression index value will help us in settlement analysis. If the natural moisture content of soil is closer to liquid limit, the soil can be considered as soft if the moisture content is lesser than liquid limit the soil is brittle and stiffer. These tests are mainly used on clayey or silty soils since these are the soils that expand and shrink due to moisture content. Clays and silts react with the water and thus change sizes and have varying shear strengths. Thus these tests are used widely in the preliminary stages of designing any structure to ensure that the soil will have the correct amount of shear strength and not too much change in volume as it expands and shrinks with different moisture contents.

THEORY

Plastic limit

The plastic limit (PL) is the water content where soil starts to exhibit plastic behavior. A thread of soil is at its plastic limit when it is rolled to a diameter of 3 mm or begins to crumble. To improve consistency, a 3 mm diameter rod is often used to gauge the thickness of the thread when conducting the test.

Liquid limit

The Liquid limit (LL) is defined as the arbitrary limit of water content at which the soil is just about to pass from the plastic state into the liquid state. At this limit, the soil possesses a small value of shear strength, losing its ability to flow as a liquid. In other words, the liquid limit is the minimum moisture content at which the soil tends to flow as a liquid.

Plasticity index

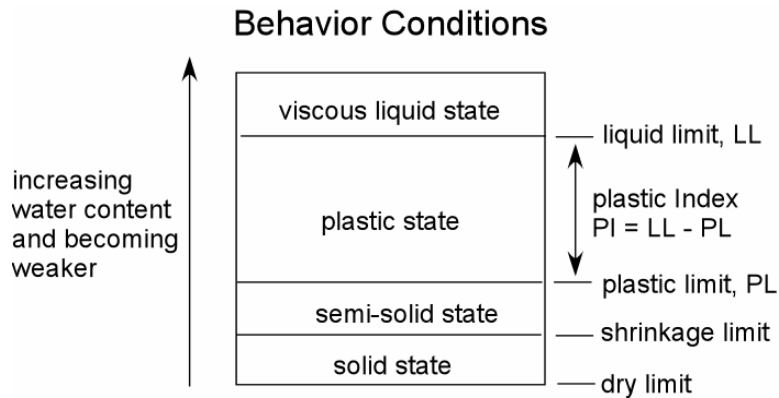
The Plasticity index (PI) is the range of water content within which the soil exhibits plastic properties; that is, it is the difference between liquid and plastic limits.

$$PI = (LL - PL)$$

When the plastic limit cannot be determined, the material is said to be non-plastic (NP). Plasticity index for sands is zero.

For proper evaluation of the plasticity properties of a soil, it has been found desirable to use both the liquid limit and the plasticity index values.

Atterberg limits are the limits of water content used to define soil behavior. The consistency of soils according to Atterberg limits gives the following diagram.



APPARATUS REQUIRED

1. For Liquid limit determination

The apparatus required are the mechanical liquid limit device, grooving tool, porcelain evaporating dish, flat glass plate, spatula, palette knives, balance, oven wash bottle with distilled water and containers.

2. For Plastic limit determination

The apparatus consists of a porcelain evaporating dish, about 12 cm in diameter (or a flat glass plate, 10 mm thick and about 45 cm square), spatula, about 8 cm long and 2 cm wide (or palette knives, with the blade about 20 cm long and 3 cm wide, for use with flat glass plate for mixing soil and water), a ground-glass plate, about 20×15 cm, for a surface for rolling, balance, oven, containers, and a rod, 3 mm in diameter and about 10 cm long.

Liquid limit device (Casagrande apparatus)

It consists of a brass dish, handle, and cam mounted on a hard rubber base (Fig. 1). The dish falls through a distance of 1 cm per rotation. A sample of soil 1 cm thick is placed in the dish with a groove 11 mm wide at the top and 2 mm at the bottom. The number of jars required to cause the 2-mm gap to close along 1/2 in (12.7 mm) is recorded.

TEST PROCEDURE

Liquid Limit:

- (1) Take roughly 3/4 of the soil and place it into the porcelain dish. Thoroughly mix the soil with a small amount of water until it appears as a smooth uniform paste. Cover the dish with cellophane to prevent moisture from escaping.
- (2) Weight the empty moisture cans with their lids, and record the respective weights.
- (3) Adjust the liquid limit apparatus by checking the height of drop of the cup. The point on the cup that comes in contact with the base should rise to a height of 10 mm. The block on the end of the grooving tool is 10 mm high and should be used as a gage. Practice using the cup and determine the correct rate to rotate the crank so that the cup drops approximately two times per second.

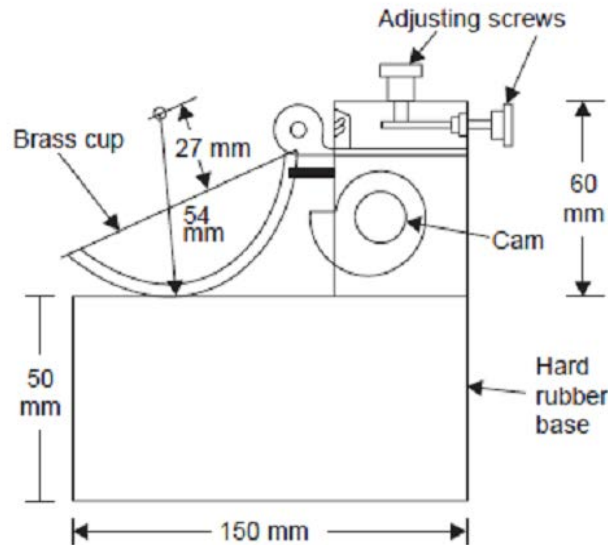


Fig. 1: Liquid limit device

(4) Place a portion of the previously mixed soil into the cup of the liquid limit apparatus at the point where the cup rests on the base. Squeeze the soil down to eliminate air pockets and spread it into the cup to a depth of about 10 mm at its deepest point. The soil pat should form an approximately horizontal surface (See Photo B).

(5) Use the grooving tool carefully cut a clean straight groove down the center of the cup. The tool should remain perpendicular to the surface of the cup as groove is being made. Use extreme care to prevent sliding the soil relative to the surface of the cup (See Photo C).

(6) Make sure that the base of the apparatus below the cup and the underside of the cup is clean of soil. Turn the crank of the apparatus at a rate of approximately two drops per second and count the number of drops, N , it takes to make the two halves of the soil pat come into contact at the bottom of the groove along a distance of 13 mm (1/2 in.) (See Photo D). If the number of drops exceeds 50, then go directly to step eight and do not record the number of drops, otherwise, record the number of drops.

(7) Take a sample, using the spatula, from edge to edge of the soil pat. The sample should include the soil on both sides of where the groove came into contact. Place the soil into a moisture can cover it. Immediately weigh the moisture can containing the soil, record its mass, remove the lid, and place the can into the oven. Leave the moisture can in the oven for at least 16 hours. Place the soil remaining in the cup into the porcelain dish. Clean and dry the cup on the apparatus and the grooving tool.

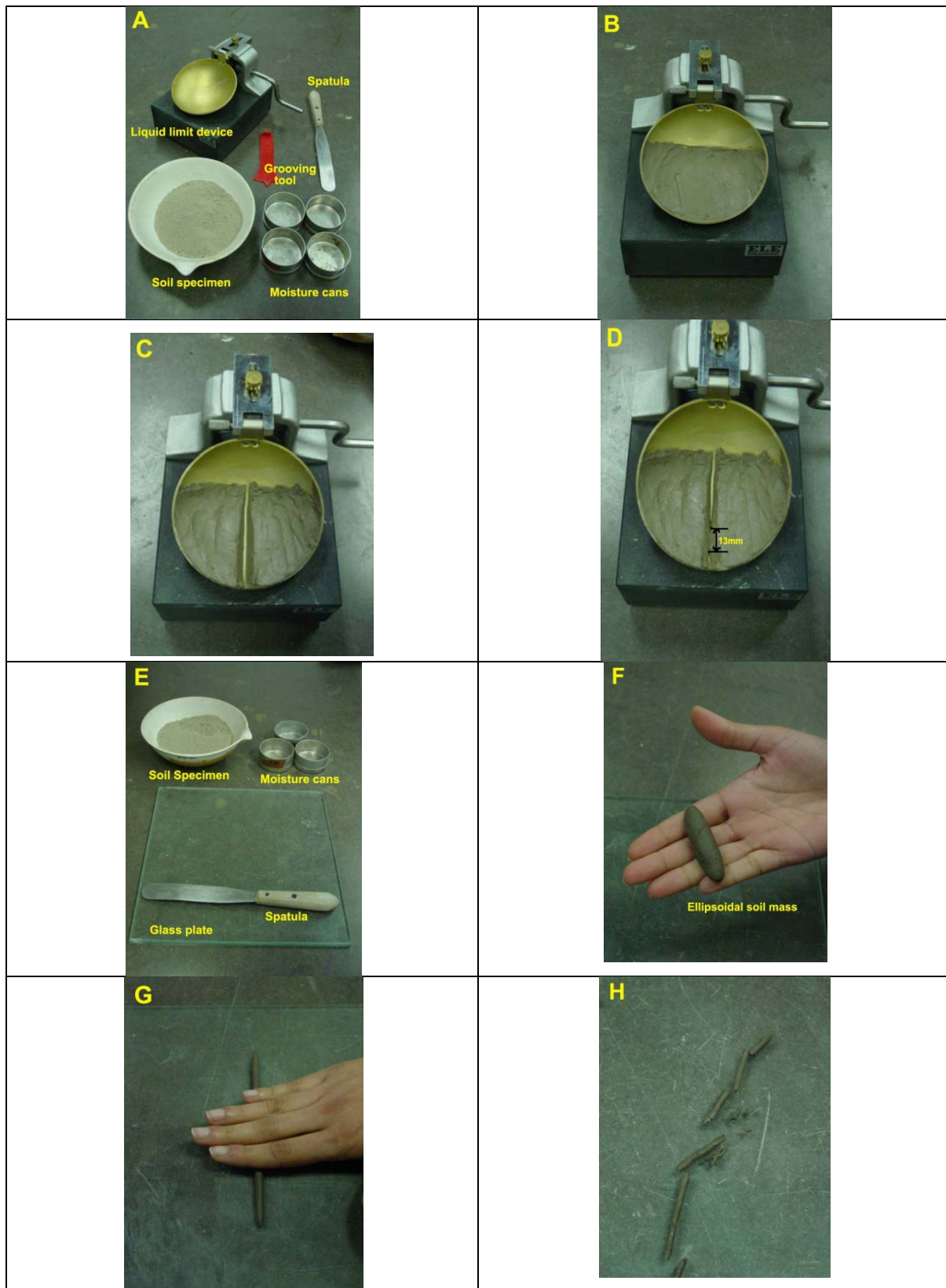
(8) Remix the entire soil specimen in the porcelain dish. Add a small amount of water to increase the water content so that the number of drops required to close the groove decrease.

Plastic Limit:

(1) Weight the remaining empty moisture can with their lids, and record the respective weights.

(2) Take the remaining 1/4 of the original soil sample and add water until the soil is at a consistency where it can be rolled without sticking to the hands.

(3) Form the soil into an ellipsoidal mass (See Photo F). Roll the mass between the palm or the fingers and the glass plate (See Photo G). Use sufficient pressure to roll the mass into a thread of uniform diameter by using about 90 strokes per minute. (A stroke is one complete motion of the hand forward and back to the starting position.) The thread shall be deformed so that its diameter reaches 3.2 mm (1/8 in.), taking no more than two minutes.



(4) When the diameter of the thread reaches the correct diameter, break the thread into several pieces. Knead and reform the pieces into ellipsoidal masses and re-roll them. Continue this alternate rolling, gathering together, kneading and re-rolling until the thread crumbles under

the pressure required for rolling and can no longer be rolled into a 3.2 mm diameter thread (See Photo H).

(5) Gather the portions of the crumbled thread together and place the soil into moisture can, then cover it. If the can does not contain at least 6 grams of soil, add soil to the can from the next trial (See Step 6). Immediately weigh the moisture can containing the soil, record its mass, remove the lid, and place the can into the oven. Leave the moisture can in the oven for at least 16 hours.

ANALYSIS

Liquid Limit

(1) Calculate the water content of each of the liquid limit moisture cans after they have been in the oven for at least 16 hours.

(2) Plot the number of drops, N , (on the log scale) versus the water content (w). Draw the best-fit straight line through the plotted points and determine the liquid limit (LL) as the water content at 25 drops.

Plastic Limit

(1) Calculate the water content of each of the plastic limit moisture cans after they have been in the oven for at least 16 hours.

(2) Compute the average of the water contents to determine the plastic limit, PL.

(3) Calculate the plasticity index, $PI = LL - PL$.

Report the liquid limit, plastic limit, and plasticity index to the nearest whole number.

Examples of Calculations

Sample Description: Grayey silty clay

Liquid Limit Determination

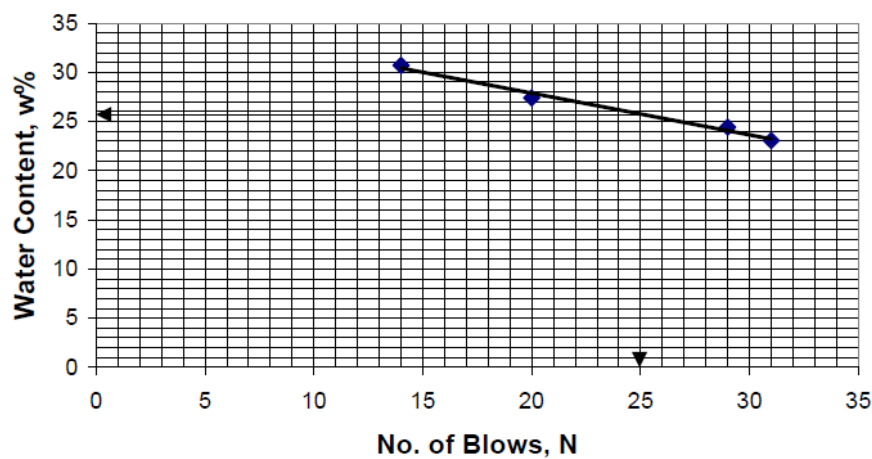
Sample No.	1	2	3	4
Moisture can and lid number	11	1	5	4
M_C = Mass of empty, clean can + lid (grams)	22.23	23.31	21.87	22.58
M_{CMS} = Mass of can, lid, and moist soil (grams)	28.56	29.27	25.73	25.22
M_{CDS} = Mass of can, lid, and dry soil (grams)	27.40	28.10	24.90	24.60
M_s = Mass of soil solids (grams)	5.03	4.79	3.03	2.02
M_w = Mass of pore water (grams)	1.16	1.17	0.83	0.62
w = Water content, $w\%$	23.06	24.43	27.39	30.69
No. of drops (N)	31	29	20	14
M_C = Mass of empty, clean can + lid (grams)	22.23	23.31	21.87	22.58

Plastic Limit Determination

Sample no.	1	2	3
Moisture can and lid number	7	14	13
M _C = Mass of empty, clean can + lid (grams)	7.78	7.83	15.16
M _{CMS} = Mass of can, lid, and moist soil (grams)	16.39	13.43	21.23
M _{CDS} = Mass of can, lid, and dry soil (grams)	15.28	12.69	20.43
M _S = Mass of soil solids (grams)	7.5	4.86	5.27
M _W = Mass of pore water (grams)	1.11	0.74	0.8
w = Water content, w%	14.8	15.2	15.1

$$\text{Plastic Limit (PL)} = \text{Average } w \% = \frac{14.8 + 15.2 + 15.1}{3} = 15.0$$

LIQUID LIMIT CHART



From the above graph, Liquid Limit = 26

Final Results:

Liquid Limit = 26;

Plastic Limit = 15, and

Plasticity Index = 11

Name of the experiment: Determination of Consistency of Soils: Shrinkage Limit Test

NEED AND SCOPE

As the soil loses moisture, either in its natural environment, or by artificial means in laboratory it changes from liquid state to plastic state to semi-solid state and then to solid state. The volume is also reduced by the decrease in water content. But, at a particular limit the moisture reduction causes no further volume change. A shrinkage limit test gives a quantitative indication of how much moisture can change before any significant volume change and to also indication of change in volume. The shrinkage limit is useful in areas where soils undergo large volume changes when going through wet and dry cycles (e.g. earth dams). This parameter is extremely important and used to understand the shrinkage and expansive property of road making materials, during construction of highways. The parameter is also important for backfilling in mines.

APPARATUS

- 1) Evaporating Dish of Porcelain,
- 2) Spatula and Straight Edge,
- 3) Balance-Sensitive to 0.01 g minimum.
- 4) Shrinkage Dish. Circular, porcelain or non-corroding metal dish,
- 5) Glass cup. 50-55 mm in diameter and 25 mm in height,
- 6) Glass plates. Two, 75X75 mm one plate of plain glass and the other with prongs,
- 7) Thermostatically controlled Oven,
- 8) Wash bottle containing distilled water,
- 9) Graduate-Glass, with capacity of 25 ml.
- 10) Mercury.

PROCEDURE

Preparation of soil paste

1. Take about 100 gm of soil sample from a thoroughly mixed portion of the material passing through 425- μ m I.S. sieve. Place about 30 gm the above soil sample in the evaporating dish and thoroughly mixed with distilled water and make a creamy paste. (Use water content slightly higher than the liquid limit.)

Filling the shrinkage dish

2. Coat the inside of the shrinkage dish with a thin layer of Vaseline to prevent the soil sticking to the dish.
3. Fill the dish in three layers by placing approximately 1/3 rd of the amount of wet soil with the help of spatula. Tap the dish gently on a firm base until the soil flows over the edges and no apparent air bubbles exist. Repeat this process for 2nd and 3rd layers also till the dish is completely filled with the wet soil. Strike off the excess soil and make the top of the dish smooth. Wipe off all the soil adhering to the outside of the dish.
4. Weigh immediately, the dish with wet soil and record the weight.
5. Air- dry the wet soil cake for 6 to 8 hrs, until the colour of the pat turns from dark to light. Then oven-dry the cake at 105°C to 110°C say about 12 to 16 hrs.
6. Remove the dried disk of the soil from oven. Cool it in a desiccator. Then obtain the weight of the dish with dry sample.
7. Determine the weight of the empty dish and record.
8. Determine the volume of shrinkage dish which is evidently equal to volume of the wet soil as follows. Place the shrinkage dish in an evaporating dish and fill the dish with mercury till it overflows slightly. Press it with plain glass plate firmly on its top to remove excess

mercury. Pour the mercury from the shrinkage dish into a measuring jar and find the shrinkage dish volume directly. Record this volume as the volume of wet soil pat.

Volume of the Dry Soil Pat

9. Determine the volume of dry soil pat by removing the pat from the shrinkage dish and immersing it in the glass cup full of mercury in the following manner.

Place the glass cup in a larger one and fill the glass cup to overflowing with mercury. Remove the excess mercury by covering the cup with glass plate with prongs and pressing it. See that no air bubbles are entrapped. Wipe out the outside of the glass cup to remove the adhering mercury. Then, place it in another larger dish, which is, clean and empty carefully. Place the dry soil pat on the mercury. It floats submerge it with the pronged glass plate which is again made flush with top of the cup. The mercury spills over into the larger plate. Pour the mercury that is displaced by the soil pat into the measuring jar and find the volume of the soil pat directly.



Fig. 1: Shrinkage Limit Test Apparatus

TABULATION AND RESULTS

S.No	Determination No.	1	2	3
1	Wt. of container in gm, W_1			
2	Wt. of container + wet soil pat in gm, W_2			
3	Wt. of container + dry soil pat in gm, W_3			
4	Wt. of oven dry soil pat, W_0 in gm			
5	Wt. of water in gm			
6	Moisture content (%), W			
7	Volume of wet soil pat (V), in cm			
8	Volume of dry soil pat (V_0) in cm ³ By mercury displacement method a. Weight of displaced mercury b. Specific gravity of the mercury			
9	Shrinkage limit $(W_s) = [W - (V - V_0) \times \gamma_w / W_0] \times 100$			
10	Shrinkage ratio (R)			

CAUTION: DO NOT TOUCH THE MERCURY WITH GOLD RINGS.

Experiment-13.B

Name of the experiment: Determination of Compaction of Soil

OBJECTIVE

The objective of this test is determination of water content-dry density relation using light compaction.

THEORY

Compaction, in general, is the densification of soil by removal of air, which requires mechanical energy. The degree of compaction of a soil is measured in terms of its dry density. The degree of compaction depends upon the moisture content, the amount of compactive effort or energy expended and the nature of the soil. A change in moisture content or compactive effort brings about a change in density. Thus, for compaction of soil, a certain amount of water and a certain predetermined amount of rolling are necessary.

The following are the important effects of compaction:

- (i) Compaction increases the dry density of the soil, thus increasing its shear strength and bearing capacity through an increase in frictional characteristics;
- (ii) Compaction decreases the tendency for settlement of soil; and,
- (iii) Compaction brings about a low permeability of the soil.

To determine the soil moisture-density relationship and to evaluate a soil as to its suitability for making fills for a specific purpose, the soil is subjected to a compaction test. Proctor (1933) showed that there exists a definite relationship between the soil moisture content and the dry density on compaction and that, for a specific amount of compaction energy used, there is a particular moisture content at which a particular soil attains its maximum dry density. Such a relationship provides a satisfactory practical approach for quality control of fill construction in the field.

The relation between moisture content and dry density of a soil at a particular compaction energy or effort is shown in Fig.1. The addition of water to a dry soil helps in bringing the solid particles together by coating them with thin films of water. At low water content, the soil is stiff and it is difficult to pack it together. As the water content is increased, water starts acting as a lubricant, the particles start coming closer due to increased workability and under a given amount of compactive effort, the soil-water-air mixture starts occupying less volume, thus effecting gradual increase in dry density. As more and more water is added, a stage is reached when the air content of the soil attains a minimum volume, thus making the dry density a maximum. The water content corresponding to this maximum dry density is called the 'optimum moisture content'. Addition of water beyond the optimum reduces the dry density because the extra water starts occupying the space which the soil could have occupied.

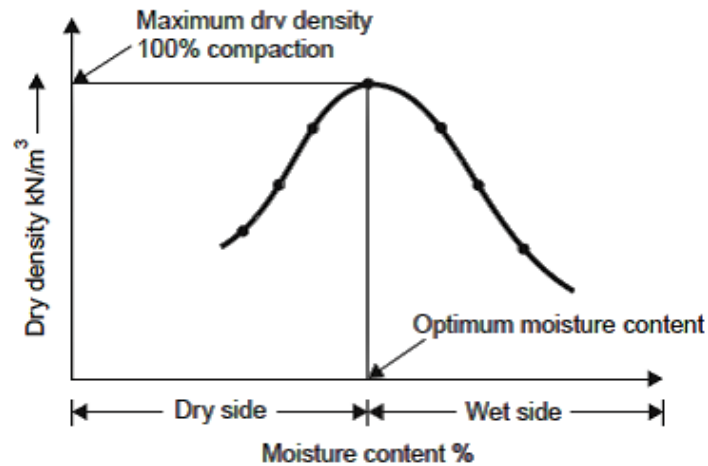


Fig. 1: Moisture content versus dry density at a particular compactive effort

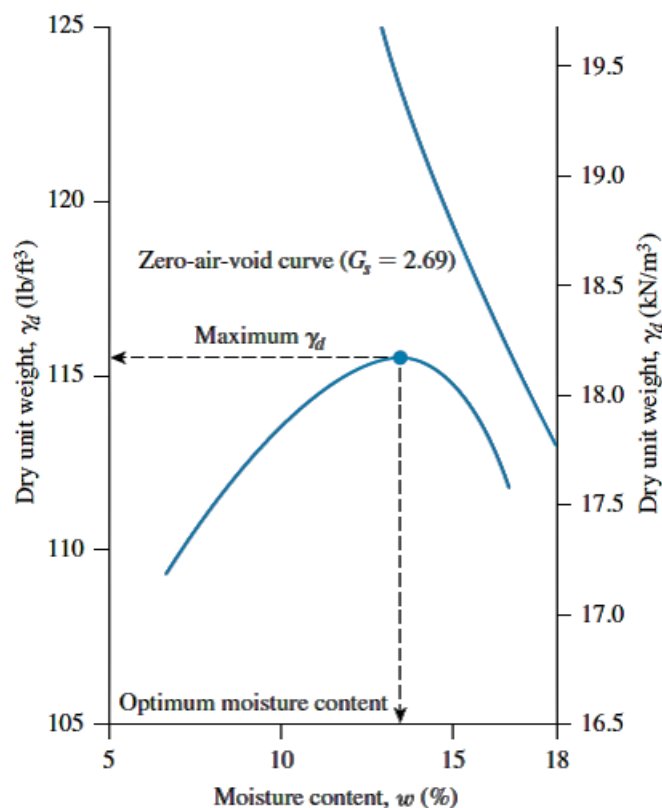


Fig.2: Standard Proctor compaction test results for a silty clay

The curve with the peak shown in Fig.1 is known as the ‘moisture-content dry density curve’ or the ‘compaction curve’. The state at the peak is said to be that of 100% compaction at the particular compactive effort; the curve is usually of a hyperbolic form, when the points obtained from tests are smoothly joined. The dry density is maximum at the optimum water content. A curve is drawn between the water content and dry density to obtain the maximum dry density and optimum water content.

Compaction method cannot remove all the air voids and therefore, the soil never becomes fully saturated. Thus the theoretical maximum dry density is only hypothetical. For given water content, theoretical maximum density is obtained corresponding to the condition when there are no air voids (i.e. degree of saturation is equal to 100%). The theoretical maximum dry

density is also known as saturated dry density. In this condition, the soil becomes saturated by reduction in air voids to zero but with no change in water content. The soil could also become saturated by increasing the water content such that all air voids are filled. The line indicating theoretical maximum dry density can be plotted along with the compaction curve.

In the construction of highway embankments, earth dams, and many other engineering structures, loose soils must be compacted to increase their unit weights. Compaction increases the strength characteristics of soils, which increase the bearing capacity of foundations constructed over them. Compaction also decreases the amount of undesirable settlement of structures and increases the stability of slopes of embankments. Smooth-wheel rollers, sheepfoot rollers, rubber-tired rollers, and vibratory rollers are generally used in the field for soil compaction. Vibratory rollers are used mostly for the densification of granular soils.

The purpose of laboratory testing is to determine the proper amount of mixing water to be used, when the compacting soil in the field and resulting degree of compactness which can be expected from compaction at optimum moisture content.

APPARATUS REQUIRED

- 1) Moulds: dimension of mould (height 127mm and inner diameter 100 mm)
- 2) Sample extruder
- 3) Weighing balances
- 4) Oven
- 5) Container: to determine water content.
- 6) Steel straightedge: 30 cm in a length and having one beveled edge.
- 7) Sieve: 4.74mm and 19 mm IS-Sieve
- 8) Mixing tools: tray or pan, spoon, trowel and spatula
- 9) Metal rammer: having mass of moving part $2.6 \text{ kg} \pm 25 \text{ gm}$ and the length of guide pipe shall be such that as to give a fall of $310 \pm 0.5 \text{ mm}$.

PROCEDURE

- a) Take a representative portion of air dried soil material and of sufficient quantity such that 6 kg of material passing through 20 mm IS sieve for soils not susceptible to crushing during compaction, or 15 kg of material passing through 19-mm IS sieve for soils susceptible to crushing during compaction.
- b) Sieve the 15 kg sample through 19-mm IS sieve and broken down sample so that, it will sieved through 4.75-mm sieve.
- c) Take five samples each of 2.5 kg and mixed each sample thoroughly with a suitable amount of water i.e. for sandy and gravelly soil, moisture content 4 to 6 % and for cohesive soil, moisture content less than 8 to 10 % below plastic limit are required.
- d) Keep the soil samples in desiccators for 16 hrs such that water is soaked uniformly.
- e) Weight the empty mould with base plate which is m_1 and measure the volume of the mould (V).
- f) Place the mould in solid place and use oil or grease in mould, collar and rammer so that soil will not attached in mould. Fill the mould with soil sample prepared in three layers and each layer is given 25 blows from 2.6 kg rammer and from height 310mm above soil. After each layer compaction, scrub surface of soil with spatula so that another layer bond together then keep another layer and give 25 blows.
- g) Remove the collar and remove the extended soil and leveled the compacted soil by using straightedge.

- h) Weight the mould with base plate and soil sample (m_2).
- i) Then remove the soil specimen from mould and take the soil of different three layers for determination water content (w).
- j) Repeat the test for at least five times such that maximum dry density will occur at range of water content.

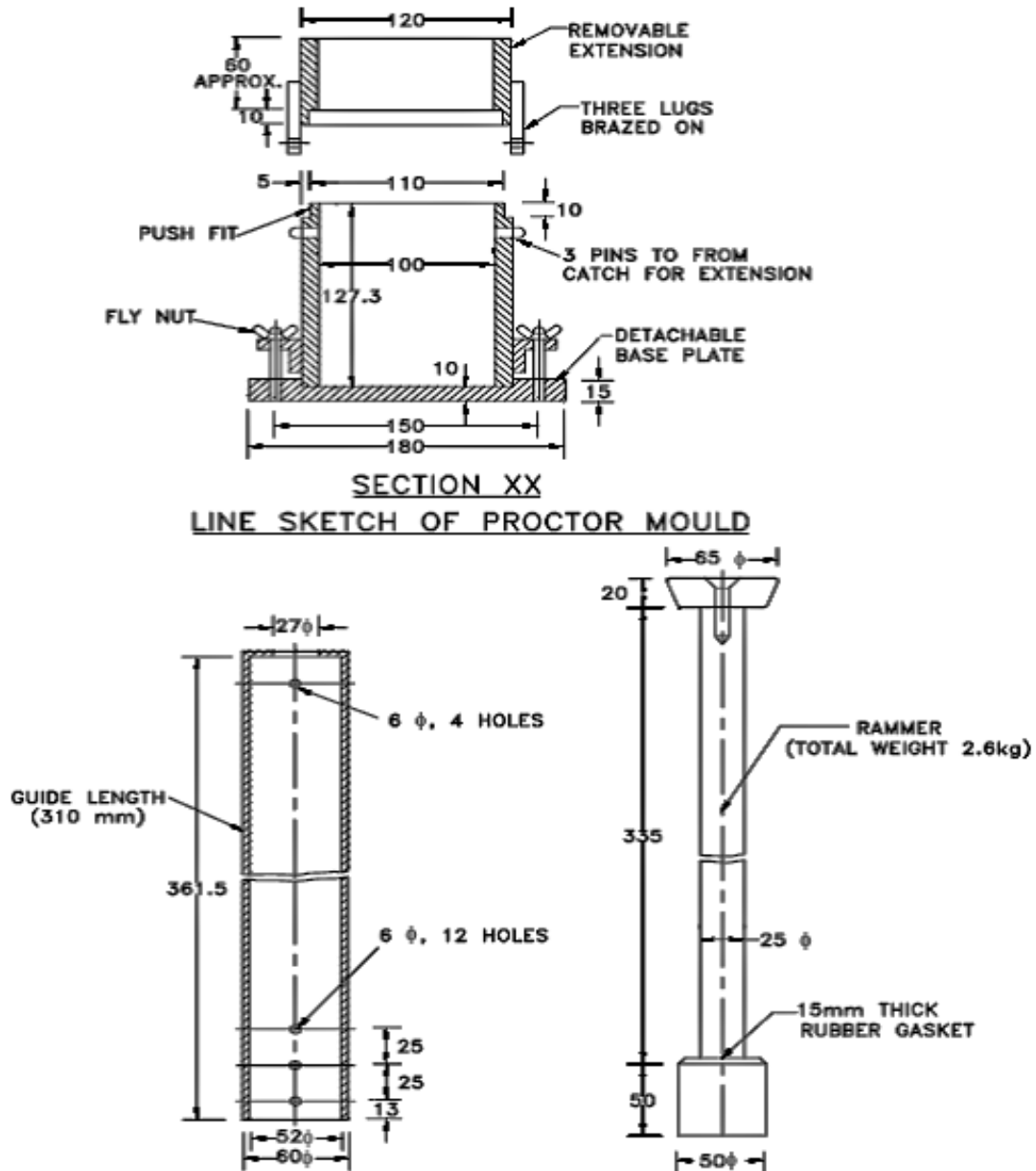


Fig. 3: Layout of metal rammer and mould according to IS



Fig. 4: Typical Proctor Compaction Test Equipment

CALCULATIONS

Bulk Density (γ_m): The bulk density (g/ml) of soil is calculated as follows:

$$\gamma_m = \frac{m_2 - m_1}{V_m}$$

Where,

m_1 = mass in g of mould and base;

m_2 = mass in g of mould, base and soil and

V_m = volume of mould

Dry Density (γ_d): The dry density (g/ml) can be calculated as:

$$\gamma_d = \frac{100 \gamma_m}{100 + w}$$

Where,

w = water content of soil in percent.

The theoretical dry density can be calculated from:

$$\gamma_{d_{theomax}} = \frac{G \gamma_w}{1 + wG}$$

Where,

G = specific gravity of soil

γ_w = density of water

The dry densities, γ_d obtained in a series of determinations shall be plotted against the corresponding moisture contents w . A smooth curve shall be drawn through the resulting points and the position of the maximum on this curve shall be determined.

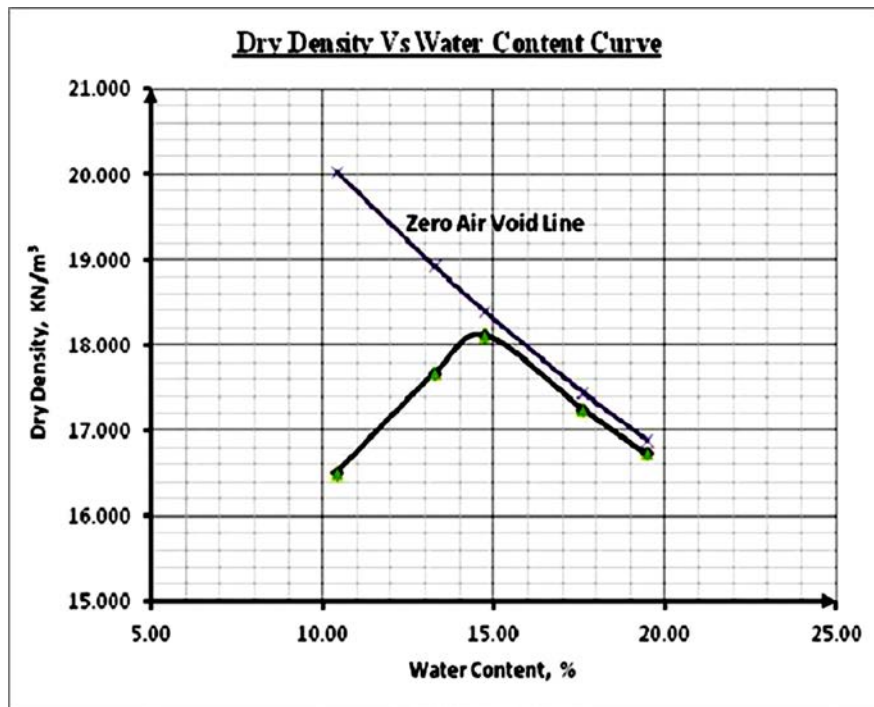
REPORTING OF RESULTS

- 1) The experimental points and the smooth curve drawn through them showing the relationship between moisture content and dry density shall be reported.
- 2) The dry density in g/ml corresponding to the maximum point on the moisture content/dry density curve shall be reported as the maximum dry density to the nearest 0.01.
- 3) The percentage moisture content corresponding to the maximum dry density on the moisture content/dry density curve shall be reported as the optimum moisture content and quoted to the nearest 0.2 for values below 5 percent to the nearest 0.5 for values from 5 percent to 10 percent, and to the nearest whole number for value exceeding 10 percent.
- 4) The amount of stone retained on the 20-mm IS Sieve shall be reported to the nearest 1 percent.
- 5) The method of obtaining the result shall be stated, (2.6-kg rammer method). The procedure used shall also be stated that is single sample or separate sample and the size of the mould used.

TEST RESULTS – A CASE STUDY

OBSERVATIONS & CALCULATIONS

DETERMINATION OF MAXIMUM DRY DENSITY AND OPTIMUM WATER CONTENT													
Name of Test: Light Compaction													
Description of Soil: Red clay													
Wt. of base plate and Mould =			4.006	kg				Specific Gravity (G) =		2.59			
Height of Mould (Ht) =			127	mm									
Diameter of Mould =			100	mm									
Volume of the Mould =			0.000997	m ³									
S.N.	Wt. of Mould with base plate + soil	Wt of soil in Mould (W), Kg.	Density of Soil (γ), KN/m ³	Can No.	Wt. of Can	Wt. of wet soil + Can	Wt. of Dry soil + Can	Wt. of Dry soil (M), kg	Wt. of Water (M _w), kg	Water Content (w), %	Average water Content (w),%	Dry Density (γ _d), kN/m ³	Dry Density At Zero void, (γ _d), kN/m ³
1	5.825	1.819	18.236	c-1	9.40	18.62	17.74	8.34	0.88	10.55	10.39	16.521	20.02
				2	10.64	16.41	15.88	5.24	0.53	10.11			
				c-2	9.97	19.66	18.74	8.77	0.92	10.49			
2	6.003	1.997	20.021	5	12.03	22.03	20.87	8.84	1.16	13.12	13.24	17.680	18.92
				c-3	10.99	23.69	22.18	11.19	1.51	13.49			
				4	9.37	18.60	17.53	8.16	1.07	13.11			
3	6.08	2.074	20.793	11	6.33	27.19	24.49	18.16	2.70	14.87	14.73	18.123	18.39
				8	15.31	29.44	27.66	12.35	1.78	14.41			
				3	8.48	24.74	22.63	14.15	2.11	14.91			
4	6.03	2.024	20.292	6	8.59	25.04	22.59	14.00	2.45	17.50	17.59	17.256	17.45
				10	11.46	41.54	37.00	25.54	4.54	17.78			
				9	11.00	27.51	25.05	14.05	2.46	17.51			
5	6.00	1.994	19.991	1	9.38	27.19	24.31	14.93	2.88	19.29	19.45	16.736	16.90
				7	11.98	23.04	21.25	9.27	1.79	19.31			
				12	11.10	26.74	24.16	13.06	2.58	19.75			



RESULT AND DISCUSSION:

The compaction characteristic curve obtained from light compaction is shown in above figure. From figure it is observed that optimum moisture content (OMC) is 14.5% and maximum dry density (γ_d)_{max} is 18.9 kN/m³. But practically it is not possible to remove 100% air void in the field from this OMC. Because we can see in the graph that zero air void line doesn't touch the compaction curve, means the density obtained in laboratory is lesser than the density at zero air voids.

INFERENCE

The optimum moisture content and dry density of soil are used to measure the field compaction by calculating relative compaction. OMC and dry density of soil also affect in shear strength, soil structure, permeability, void ratio etc. Hence, these values are necessary for different civil construction work like highway, dam, embankment etc.

REFERENCES

IS: 2720 (Part 7), 1987, Methods of test for soils: Part 7 - Determination of water content-dry density relation using light compaction (second revision).